

Diagnostic Interpretation of Effective Latent Dimensionality in Autoencoders and Variational Autoencoders

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Abstract

This paper proposes a multi-indicator diagnostic framework that integrates active-unit (AU) dynamics, intrinsic-dimension (ID) estimation (TwoNN/MLE), reconstruction-error (MSE) elbows, and PCA baselines to diagnostically interpret the effective latent dimensionality actually used in the representations learned by autoencoders (AEs) and variational autoencoders (VAEs). Rather than determining the effective latent dimensionality as a single unique number, the framework explicitly separates the diagnostic roles of the indicators: AU values serve as robust direct observations of the number of KL-active latent coordinates; TwoNN/MLE ID estimates serve as numerically stable auxiliary reference estimators (stable latent-space references whose unbiasedness with respect to the true intrinsic dimension is not guaranteed); and the AU slowdown point m^{knee} is treated as a TwoNN-guided exploratory change point. Theoretically, under a rate–distortion surrogate of the linear-Gaussian β -VAE—an interpretive surrogate, not a theorem about trained nonlinear VAEs—we derive the condition $\lambda_j > \beta\tau^2$ for the j -th eigendirection of the data covariance to be activated. Experimentally, we show that in fully connected VAEs trained on MNIST, AU values and TwoNN IDs are highly reproducible across random seeds, whereas m^{knee} fluctuates strongly on sparse m -grids, and that a dense grid search around the TwoNN estimate (TwoNN-guided local refinement) stabilizes m^{knee} at 10.3 ± 0.9 —with the scope of this stabilization limited to MNIST FC-VAEs with $\beta = 4$, a dense grid, and three seeds. Furthermore, through Fashion-MNIST, Conv-VAE, and CIFAR-10/SVHN experiments, we quantify the conditions under which the framework functions and the boundary conditions under which AU/ m^{knee} diagnostics become inconclusive.

Keywords: variational autoencoder; effective latent dimensionality; active units; intrinsic dimension; TwoNN; diagnostic framework; representation learning; knowledge extraction; posterior collapse

1. Introduction

1.1. Background: The Problem of Interpreting Effective Latent Dimensionality

In representation learning, autoencoders (AEs) and variational autoencoders (VAEs) [12] are core unsupervised models that learn low-dimensional latent representations from high-dimensional data [7]. The most fundamental design parameter governing the “quality” of these latent representations is the bottleneck dimension m : if m is too small, information is lost, whereas if it is too large, redundant dimensions arise (and, in VAEs, posterior collapse worsens [21]).

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However, a methodology for quantitatively reading “how many dimensions a learned latent representation actually uses”—that is, the **effective latent dimensionality**—has not been sufficiently established in deep representation learning research. Existing studies [17,19] have measured the intrinsic dimension of individual layers of deep networks, but studies that systematically analyze the effective dimensionality at the bottleneck of AEs/VAEs together with its training dynamics remain limited.

The central question addressed in this paper is:

How do AEs/VAEs “learn” the local intrinsic dimensionality and the topological embedding dimensionality of the input data as the effective dimensionality of the latent space? And is there a way to quantitatively read this effective dimensionality from the outside?

1.2. Positioning and Contributions of This Study

To address this question, we propose a **multi-indicator diagnostic framework** built on two pillars: **intrinsic-dimension (ID) estimation** and **active-unit (AU) dynamics**. The novelty of the framework lies not in “uniquely determining” the effective dimensionality with a single indicator, but in combining the indicators while making the robustness and exploratory character of each explicit, and **interpreting them diagnostically**. We do not claim that “ m^{knee} uniquely identifies an appropriate bottleneck dimension”— m^{knee} is a TwoNN-guided exploratory change point, and the primary readout of effective dimensionality is carried by the robust indicator (AU values) and the numerically stable auxiliary reference estimator (TwoNN ID) (the classification and reliability of the indicators are summarized in Table 2).

The **contributions of this paper are threefold**:

1. **Proposal of a multi-indicator diagnostic framework:** The main contribution of this study is not the proposal of m^{knee} itself, but the organization of AU values, TwoNN/MLE ID, the MSE elbow, and PCA as diagnostic indicators with distinct roles, providing a framework for interpreting the effective latent dimensionality of AEs/VAEs from multiple perspectives. We clearly distinguish AU values as a robust indicator that directly observes latent-dimension usage, TwoNN ID as a numerically stable auxiliary reference estimator, and m^{knee} as an exploratory change point obtained from the shape of the AU curve (Sections 3 and 6). Three-seed verification on MNIST confirms the high reproducibility of AU values ($\sigma \leq 0.5$) and TwoNN ID (10.03 ± 0.12), and we present $m^{\text{knee}} = 10.3 \pm 0.9$ under a dense grid placed around the TwoNN estimate, with independent initialization and multiple seeds (Exp-Mnist-Dense-300), as the main stabilization result (Section 5.3.6). However, because the search region itself is chosen from the TwoNN estimate, this result is a **TwoNN-guided local refinement**, not a validation independent of TwoNN. Its verified scope is limited to MNIST FC-VAEs with $\beta = 4$ and three seeds.
2. **Derivation of the AU activation condition based on a linear-Gaussian rate-distortion surrogate:** Under the linear-Gaussian β -VAE rate-distortion surrogate, we derive the condition $\lambda_j > \beta\tau^2$ for the j -th latent dimension to be active, using the water-filling theorem and the KKT conditions (Theorem 1, Section 4), and explain the mechanism by which KL regularization suppresses effective dimensionality from the perspective of the water-filling principle. This result is not a rigorous theorem for actual non-linear VAEs; it is an analytical result based on the linear-Gaussian rate-distortion surrogate that provides a qualitative correspondence with the non-linear experiments.
3. **Quantification of the applicability range and limiting conditions:** Targeting MNIST/Fashion-MNIST (FC-VAE), Conv-VAE, and CIFAR-10/SVHN, we empirically demonstrate the reproducibility of AU values and TwoNN ID, the grid and

seed dependence of m^{knee} , and that “ $m \gg \hat{d}_{\text{ID}}$ is the dominant condition for TwoNN stability, more important than isometry improvement” (Exp-Real-Geom, Proposition 4). Furthermore, for Conv-VAEs and natural images, we quantify the boundary conditions under which $\text{AU}/m^{\text{knee}}$ diagnostics become difficult, as five operational criteria for stable identification (Sections 6.6 and 6.7).

1.3. Two Kinds of “Dimension”: Local Intrinsic Dimension and Topological Embedding Dimension

In this paper, we distinguish two notions of manifold “dimension”: d_{ID} (the local intrinsic dimension, the target of TwoNN/MLE estimation) and d_{emb} (the topological minimal embedding dimension, the minimal dimension required for an embedding without self-intersection) (for the Swiss Roll, $d_{\text{emb}} = d_{\text{ID}} = 2$; for the Torus, $d_{\text{emb}} = 3 > d_{\text{ID}} = 2$).

Note, however, that **the main contribution of this paper is not to prove a correspondence with d_{ID} or d_{emb} , but to organize the robustness, limits, and applicability conditions of the diagnostic indicators AU values, TwoNN ID, and m^{knee} .** The conditional correspondence between d_{emb} and AU_{sat} (for the Torus, Möbius strip, etc.) is positioned as an incidental supporting observation (Section 6.5).

1.4. Research Questions

RQ1 Under what grid-density, seed-count, and architecture conditions can the change point m^{knee} obtained from the AU dynamics of a VAE be used as a diagnostic auxiliary quantity consistent with TwoNN/MLE ID estimates?

RQ2 Does a framework that interprets the effective latent dimensionality of AEs/VAEs with multiple indicators (AU values, TwoNN/MLE, the MSE elbow, and PCA) provide complementary information relative to any single indicator?

RQ3 How do bottleneck headroom (the magnitude of $m/\hat{d}_{\text{ID}}$) and geometric regularization (IsometricAE) affect the stability of TwoNN ID estimation?

1.5. Paper Organization

Explicit scope of validation: The validation in this paper covers synthetic manifolds ($d_{\text{true}} \leq 3$), MNIST and Fashion-MNIST (fully connected AE/VAE), Conv-AE/VAE, and natural images (CIFAR-10 and SVHN). In the natural-image experiments, we confirm that the TwoNN auxiliary estimate yields values consistent with those reported by Pope et al. [16], and we quantify the validity range and limiting conditions of $\text{AU}/m^{\text{knee}}$ diagnostics for Conv-VAEs (we do not claim “generalization” of the entire framework to natural images).

The paper is organized as follows: Section 2 (related work), Section 3 (framework), Section 4 (theoretical background, including Theorem 1), Section 5 (experiments), Section 6 (discussion), and Section 7 (conclusions). Detailed proofs and sensitivity analyses are collected in the appendices.

2. Related Work

2.1. Representation Learning and Autoencoders

Bengio et al. [7] organized the theoretical framework of representation learning and discussed the mechanism by which AEs learn the low-dimensional structure of high-dimensional data based on the manifold hypothesis [6]. β -VAE [20] promotes disentangled representations by increasing the KL-regularization weight β , which directly affects AU. Lucas et al. [21] analyzed, within a linear-VAE framework, the conditions under which posterior collapse (a substantial reduction of AU) occurs in ELBO minimization, and qualitatively discussed the relationship between AU and the variance structure (effective dimensionality) of the data. From the perspective of score matching, the DAE analysis [23]

showed that the reconstruction directions of denoising autoencoders selectively suppress the normal directions of the manifold. The CAE [22] learns contractive representations by regularizing the encoder-side Jacobian.

2.2. Intrinsic-Dimension Analysis of Deep Representations

Pope et al. [16] showed via TwoNN estimation that the intrinsic dimension of natural images is far lower than the pixel dimension (CIFAR-10: $d_{\text{ID}} \approx 26$ at 32×32). Ansuini et al. [17] quantified the patterns by which ID changes across the layers of deep networks, documenting contraction and expansion phenomena of the representations. Valeriani et al. [19] systematically analyzed the geometric structure of the hidden representations of large transformer models. Birdal et al. [18] analyzed the relationship to generalization error by integrating persistent homology with ID. Our study is complementary to these works that “measure the ID of existing networks,” and differs in that it tracks the formation of effective dimensionality at the **design stage** and during the **training dynamics** of AEs/VAEs.

As intrinsic-dimension estimation methods, we adopt TwoNN [8] (two-nearest-neighbor distance ratios) and MLE [9] (maximum-likelihood estimation). DANCo [15] is more precise but computationally more expensive. While UMAP [25] is widely used as a non-linear dimensionality-reduction technique for visualization, our study focuses not on visualization but on the **quantitative interpretation** of the bottleneck dimension.

2.3. Active Units and Posterior Collapse

The active-unit (AU) count [20,21] is the number of latent dimensions whose posterior-mean variance exceeds a threshold δ , and it quantifies the number of dimensions in which KL regularization is functioning effectively. Lucas et al. [21] analyzed, within a linear-VAE framework, the conditions under which ELBO minimization induces posterior collapse, and qualitatively discussed the relationship between AU and the eigenvalue structure of the data. In contrast, our Theorem 1 **quantifies Lucas et al.’s qualitative argument as the closed-form threshold** $\lambda_j > \beta\tau^2$ in terms of the data eigenvalues λ_j , the noise τ^2 , and the regularization β ; its novelty lies in carrying out a quantitative derivation under the linear-Gaussian rate–distortion surrogate, using an explicit connection to Shannon’s water-filling theorem and the KKT conditions (it also subsumes the $\beta = 1$ special case of PPCA by Tipping & Bishop [5]). Moreover, the systematic analysis of AU growth dynamics on real data—in particular the concept of m^{knee} and its correspondence with effective representational dimensionality, its limiting conditions, and stabilization strategies—is systematically validated in this study (Theorem 1, Proposition 2, Section 6.2).

2.4. Geometric Autoencoders

Nazari et al. [14] proposed geometric autoencoders, introducing a regularizer that makes the encoder preserve the geometric structure of the manifold. Kato et al. [13] proposed IsometricAE, which realizes an isometric embedding by driving the pullback metric of the decoder Jacobian toward the identity matrix. In this study, we position IsometricAE as “an auxiliary tool that improves a precondition (isometry) for the readout accuracy of effective dimensionality” (RQ3, experiment Exp-Syn-Iso).

3. Framework: Reading Out the Effective Latent Dimensionality

3.1. Problem Setup and Main Notation

We define an AE/VAE with input dimension N and bottleneck dimension m : an encoder $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$ and a decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$. For a VAE, the encoder outputs a

posterior $q_\phi(z|x) = \mathcal{N}(\mu(x), \sigma^2(x))$, and the KL regularization (with strength β) acts as an information bottleneck.

Definition of Active Units (AU):

$$\text{AU}(m) = \sum_{k=1}^m \mathbf{1}[\text{Var}_x[\mu_k(x)] > \delta], \quad \delta = 10^{-2} \quad (1)$$

where $\mu_k(x)$ is the k -th component of the posterior mean and δ is a threshold (following Lucas et al. [21]). AU is extremely robust to this threshold choice: a sensitivity analysis confirms that the AU values remain essentially unchanged for $\delta \in [10^{-3}, 10^{-1}]$ (see Appendix G).

Definition of the AU increase rate (two steps): We first define the raw increase rate $s(m_i)$ on the grid as

$$s(m_i) = \frac{\text{AU}(m_i) - \text{AU}(m_{i-1})}{m_i - m_{i-1}} \quad (2)$$

where m_{i-1} is the immediately preceding value on the grid. We then define the normalized increase rate $\rho(m_i)$ as

$$\rho(m_i) = \frac{s(m_i)}{\max_j s(m_j)} \quad (3)$$

so that $\rho(m_i) \in [0, 1]$.

Table 1 lists the **main notation**.

Table 1. Notation used throughout this paper

Symbol	Meaning	Section
m	Bottleneck dimension (design parameter)	§3
m^{cand}	Diagnostic candidate dimension (not a unique optimum)	§3
m^{sat}	AU sharp saturation point (diagnostic candidate for synthetic manifolds)	§3
m^{knee}	AU slowdown point (rough heuristic for real data)	Equation (4)
AU_{sat}	Maximum AU value over the grid	Equation (1)
$\hat{d}_{\text{ID}}$	ID estimate via TwoNN/MLE	§3
d_{true}	True intrinsic dimension of synthetic manifold (known)	§5.2
d_{emb}	Topological minimum embedding dimension ($d_{\text{emb}} \geq d_{\text{ID}}$)	§1.3
κ	Decoder Jacobian condition number (isometry indicator)	§2.4
$s(m_i)$	Raw AU increase rate per unit dimension	Equation (2)
$\rho(m_i)$	Normalized AU increase rate ($\in [0, 1]$)	Equation (3)
τ_{knee}	Slowdown detection threshold (default: 0.25)	Equation (4)

The core of our framework is the explicit classification of each indicator as a **robust indicator** or an **exploratory indicator**, followed by their complementary use (Table 2). This classification is an empirical one grounded in the experimental findings reported later (§5.3.4), and it forms the core of a design philosophy that systematically avoids the misdiagnoses caused by overreliance on any single indicator.

On the meaning of “robust indicator” and “exploratory indicator”: “Robust” here does not mean unbiased with respect to the true intrinsic dimension. We use the term in the sense that, at least under our experimental conditions, the variance with respect to random seeds is small. “Exploratory,” in contrast, means that the indicator is useful for grid design and hypothesis generation but is not treated as a unique estimate of the effective latent dimensionality.

Table 2. Classification and empirical reliability of diagnostic indicators

Indicator	Category	Empirical reliability (inter-seed σ)	Primary use
AU values ($AU(m)$)	Robust indicator	$\sigma \leq 0.5$ (MNIST 300ep)	Direct observation of effective dimensionality; independent of reference AE geometry (depends on β , δ , model capacity)
TwoNN ID ($\hat{d}_{ID}$)	Numerically stable auxiliary reference[†]	$\sigma \leq 0.12$ ($m = 64$)	Reference estimate of intrinsic dimension
MSE elbow	Auxiliary indicator	Qualitative only	Reconstruction-side verification
PCA cumulative variance	Auxiliary indicator	Linear assumption only	Linear baseline
m^{knee} (AU slowdown point)	Exploratory indicator	$\sigma = 15$ (sparse grid); $\sigma = 0.9$ (dense grid)	Rough heuristic of effective dimensionality

[†] TwoNN ID is a **supplementary reference estimator** subject to isometry bias ($\kappa \approx 10^9$); see §3.2 for details. σ of m^{knee} is for sparse grid; improved to $\sigma = 0.9$ with dense grid (Exp-Mnist-Dense), where grid design is a major factor (§5.3.6).

Distinguishing the Target Quantities of Each Indicator

Under the umbrella term “effective latent dimensionality,” several quantities of genuinely different nature are in fact being measured. To avoid misdiagnosis, we explicitly distinguish the following five target quantities throughout this paper:

1. **Intrinsic dimension of the data manifold** d_{ID}^{data} : the local degrees of freedom of the input distribution itself. Known for synthetic manifolds (d_{true}); approximated for real data by applying TwoNN/MLE in input space.
2. **Intrinsic dimension of the latent representation** d_{ID}^{latent} : the estimate obtained by applying TwoNN/MLE to the latent representation of a reference AE (our $\hat{d}_{ID}$). Numerically stable, but mediated by the distortion of the reference-AE embedding, so agreement with (i) is not guaranteed—it is a stable **latent-space reference** (Section 6.3).
3. **Number of KL-active coordinates** (AU values): the number of latent coordinates the VAE actually uses for encoding. An **observable of coordinate usage** that depends on β , the architecture, the threshold δ , and the training conditions—not an intrinsic-dimension estimator. m^{knee} is a change point of the curve $AU(m)$, and AU_{sat} is the maximum AU value over the grid.
4. **Bottleneck dimension required for reconstruction**: the smallest m at which the reconstruction error reaches an acceptable level (captured by the MSE elbow).
5. **Diagnostic design dimension** m^{cand} : the candidate value chosen by the designer by cross-checking the indicators of (i)–(iv). Not a unique optimum.

The claims of this paper are stated on top of this distinction: Theorem 1 gives the threshold for (iii) in a linear surrogate; Proposition 4 and experiment Exp-Real-Geom address the stability conditions of (ii); and the m^{knee} stabilization result (Section 5.3.6) identifies conditions under which the change point of (iii) stabilizes near (ii). We make no claims about (i) itself except for synthetic manifolds with known d_{true} .

3.2. Pipeline for Reading Out the Effective Latent Dimensionality

The framework proposed in this study is an interpretation pipeline that “reads out the effective latent dimensionality learned by an AE/VAE in two stages” (Figure 1).

Stage 1: Approximate estimation of the intrinsic dimension (auxiliary role)

In this paper, we estimate the intrinsic dimension $\hat{d}_{ID}$ in two ways: (a) for **synthetic manifolds**, TwoNN/MLE is applied directly in the input space (allowing comparison with

d_{true}); (b) for **real data (MNIST, etc.)**, to avoid the curse of high dimensionality, the estimate is obtained via the latent representation of a larger reference AE ($m_{\text{ref}} = 256$).

Explicit statement of limitations: The latent space of the reference AE in method (b) ($m = 256 \gg \hat{d}_{\text{ID}}$) does not guarantee isometry (auxiliary measurement: $\kappa \approx 10^9$; §6.7), so the resulting $\hat{d}_{\text{ID}}$ is an approximation subject to isometry bias. In this paper, we treat $\hat{d}_{\text{ID}}$ as **auxiliary prior information** indicating the candidate range for m^{knee} , and by design we entrust the primary readout of the effective dimensionality to the AU dynamics of Stage 2. This avoids circular reasoning stemming from the non-isometry of the reference AE. We ensure practical stability by confirming that the estimates converge across multiple values of m (e.g., $\{64, 128, 256\}$).

Stage 2: Reading out the effective dimensionality via AU dynamics

We sweep the bottleneck dimension m over the discrete grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256\}$, training a VAE ($\beta = 4$) at each m and computing $\text{AU}(m)$ (some experiments use extended grids that omit or add some of the m points of $\mathcal{M}$; these are stated explicitly in each experiment section). Depending on the AU pattern, we read the diagnostic candidate dimension m^{cand} in one of two ways (m^{cand} **is not the unique “optimal” bottleneck dimension but a diagnostic candidate dimension based on multiple indicators**):

Saturation type (simple data such as synthetic manifolds) When AU saturates sharply at a specific m , we use that saturation point m^{sat} as the diagnostic candidate dimension $m^{\text{cand}} = m^{\text{sat}}$ (since AU values are robust, this is quite reliable under these conditions).

Gradual-increase type (real data, e.g., MNIST) When AU increases gradually with m , we refer to the slowdown point m^{knee} as a rough design guideline (an exploratory indicator):

$$m^{\text{knee}} = \min\{m_i \in \mathcal{M} : \rho(m_i) < \tau_{\text{knee}}\} \quad (4)$$

with $\tau_{\text{knee}} = 0.25$ (default; see Appendix C). m^{knee} roughly captures “the point at which the VAE first performs significant dimensional compression,” but its value depends on the m grid and on the random seed, so the final judgment is made by cross-checking against the AU values and the TwoNN ID (§6.1). Even if m^{knee} and $\hat{d}_{\text{ID}}$ do not agree, the framework as a whole is not invalidated.

CKA [24] and the reconstruction-error (MSE) elbow serve as auxiliary criteria, and the overall judgment is made from multiple indicators.

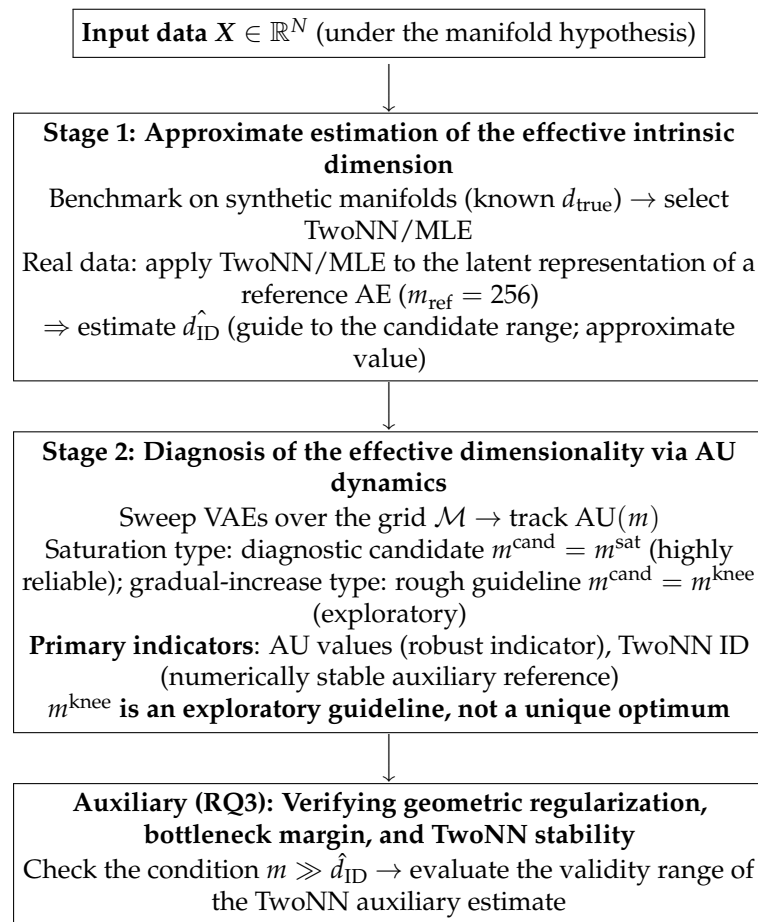


Figure 1. Diagnostic pipeline for effective latent dimensionality. The diagnostic candidate dimension m^{cand} is not a unique optimal solution but a candidate based on multiple indicators (AU values, TwoNN ID, m^{knee}).

3.3. Interpretive Significance of m^{knee}

We now describe the mechanism by which m^{knee} functions as an indicator of the effective representation dimensionality.

In the course of minimizing the VAE’s ELBO, the KL regularization (the information bottleneck term) deactivates surplus latent dimensions that are not needed for reconstruction (i.e., it induces posterior collapse) [20,21]. In the regime $m < \hat{d}_{\text{ID}}$, the bottleneck impedes reconstruction, so the VAE activates all available dimensions as AU ($\text{AU}(m) \approx m$). Once m exceeds roughly $\hat{d}_{\text{ID}}$, surplus dimensions begin to appear and the KL regularization exerts pressure to deactivate them, so the AU increase rate slows down. This slowdown point m^{knee} tends to agree with the data’s effective intrinsic dimension $\hat{d}_{\text{ID}}$ under specific conditions (e.g., Fashion-MNIST), but its numerical reproducibility depends on the m grid and on random seeds and is not guaranteed (§5.3.4, §6.1). As a framework, regardless of whether m^{knee} agrees, the readout of the effective dimensionality through the combination of AU values (a robust, directly observed quantity) and the TwoNN ID (a numerically stable auxiliary reference estimator) constitutes the central insight of this framework.

3.4. Criteria for Stable Identification

We define here, ahead of the experiments, the criteria required to claim that $\text{AU}/m^{\text{knee}}$ has been “stably identified” on a new dataset or architecture. In this paper, we say that **stable identification** holds when all five of the following conditions are met:

1. **Seed stability:** over 3–5 seeds, the standard deviation of m^{knee} satisfies $\sigma_{\text{seed}} \leq 0.1 \cdot m^{\text{knee}}$.

2. **Grid stability:** doubling the grid density changes m^{knee} by at most 20%. 285
3. **β stability:** changing $\beta_{\max}$ by $\pm 20\%$ keeps m^{knee} within the same grid cell. 286
4. **AU-plateau stability:** the across-seed standard deviation of the AU plateau value satisfies $\sigma \leq 0.05 \cdot \text{AU}_{\text{sat}}$. 287
5. **TwoNN–AU consistency:** $m^{\text{knee}} \leq 3\hat{d}_{\text{ID}}$ (coarse agreement between TwoNN and AU). 289

These thresholds ($0.1 \cdot m^{\text{knee}}$, 20%, $0.05 \cdot \text{AU}_{\text{sat}}$, $3\hat{d}_{\text{ID}}$) are not theoretically derived universal constants but **author-defined operational criteria**. The numerical choices take the scale of the across-seed variability observed for MNIST FC-VAEs (AU: $\sigma \leq 0.5$; dense-grid m^{knee} : $\sigma = 0.9 \approx 0.09 m^{\text{knee}}$; Section 5.3.6) as the yardstick for “stable,” rounded so that variability exceeding it by orders of magnitude is judged “unstable.” A systematic sensitivity analysis of the thresholds themselves is left as future work, but the conclusion of this paper (none of the five criteria are met for natural images; Section 5.11) is clear-cut enough that it does not change even if the thresholds are relaxed by a factor of two. The purpose of the criteria is not to declare “success” but to decompose and report **which conditions are violated** when stable identification is not achieved. 290–299

4. Theoretical Background 300

In this section, we present the theoretical foundations that support the framework. Proposition 1 is a lower bound based on standard topological facts about topological dimension and embedding dimension; its rigorous claim is restricted to the idealized premise of “the dimension required to achieve zero reconstruction error.” Theorem 1 is an analytical result under the linear-Gaussian rate–distortion surrogate and gives the AU activation threshold in closed form (proof details in A). Proposition 2 is a lemma under an idealized monotone-saturation model; note that in practical non-linear settings, errors exceeding the upper bound can occur. Proposition 4 is not a guarantee of unbiasedness but a sketch of sufficient conditions for conditional stability. Proposition 3 follows directly from the definition of isometry. 301–310

Proposition 1 (Topological Lower Bound on the Bottleneck Dimension). *For a smooth d -dimensional manifold $\mathcal{M} \subset \mathbb{R}^N$ to be reconstructed by a continuous AE without self-intersection, ideally achieving zero reconstruction error, a bottleneck dimension $m \geq d$ is necessary. (Practical settings involving finite samples, finite capacity, and approximate reconstruction are outside the scope of this idealization.)* 311–315

Proof. Zero reconstruction error requires $g \circ f|_{\mathcal{M}}$ to be close to the identity map, so $f|_{\mathcal{M}} : \mathcal{M} \rightarrow \mathbb{R}^m$ must be a continuous injection. By standard properties of the topological embedding dimension (Hatcher [10], Chapter 2), no continuous injection exists from a d -dimensional manifold into a Euclidean space of dimension $m < d$; hence $m \geq d$ is a necessary condition. $\square$ 316–320

This proposition justifies $\hat{d}_{\text{ID}}$ as a reference point for a principled lower bound on the bottleneck dimension (in practice, we use it as a “candidate range for the effective dimensionality”). Experiment Exp-Syn-1 confirms this: at the transition $m = 1 \rightarrow 2$, the MSE improves by roughly a factor of 35 on the Swiss Roll (Section 5.2). 321–324

Remark 1 (Relation to Prior Work). *Our theorem is closely related to a family of known analyses—the discussion of posterior collapse in linear VAEs by Lucas et al. [21], probabilistic PCA by Tipping & Bishop [5] (corresponding to the special case $\beta = 1$), and Shannon’s water-filling theorem [1]—and its novelty is limited to organizing these into a closed-form threshold $\lambda_j > \beta\tau^2$ for the β -ELBO with $\beta \neq 1$, together with monotone saturation. The role of the theorem is not to present a new* 325–329

mathematical result, but to provide a quantitative reference point for interpreting the non-linear experiments.

Theorem 1 (AU Activation Threshold under the Linear-Gaussian Rate-Distortion Surrogate). For a β -VAE with linear encoder, linear decoder, and Gaussian prior (detailed setting in A), under the linear-Gaussian rate-distortion surrogate, the water-filling theorem and the KKT conditions yield the following:

1. **(Activation threshold)** The j -th eigen-direction becomes active iff

$$\lambda_j > \beta\tau^2 \quad (5)$$

where λ_j is the j -th eigenvalue of the covariance matrix of the centered input data, $\Sigma = \mathbb{E}[(x - \bar{x})(x - \bar{x})^\top]$ (i.e., the variance of the j -th principal component in the input-space PCA basis), and τ^2 is the isotropic Gaussian observation-noise variance of the decoder output.

2. **(Upper bound on the optimal number of active eigen-directions)** The number of active eigen-directions at the global optimum, $A^*(m)$, satisfies $A^*(m) \leq \min(m, d^*(\beta))$, where $d^*(\beta) := |\{j : \lambda_j > \beta\tau^2\}|$.
3. **(Monotone saturation at the optimum)** $A^*(m)$ is monotone non-decreasing in m and saturates for $m \geq d^*(\beta)$ (i.e., $A^*(m) = d^*(\beta)$).

A detailed derivation is given in B. The quantity $\beta\tau^2$ acts as the “water level” of water-filling: only eigen-directions whose eigenvalues exceed it become active. This theorem is not a theorem about trained non-linear VAEs; it is an **interpretive surrogate** for interpreting the non-linear experiments.

Remark 2 (Relation to Empirical AU). Item (3) of Theorem 1 establishes the monotonicity of the number of active eigen-directions $A^*(m)$ at the global optimum. In contrast, practical non-linear VAEs are trained independently for each m by stochastic optimization, so the observed empirical $AU(m)$ can be **non-monotone** due to finite-sample effects, initialization dependence, and optimization noise (e.g., in Table 9, $AU = 26$ at $m = 128$ falls below $AU = 29$ at $m = 64$). Therefore, the empirical m^{knee} must be treated not as a rigorous estimate of the saturation point of A^* , but as an exploratory diagnostic indicator derived from the shape of the AU curve.

Proof sketch (details in A). Analyzing the β -ELBO contribution of each PCA direction j independently (a standard result for linear-Gaussian models), the ELBO gain ΔL_j from activating direction j is positive only when $\lambda_j > \beta\tau^2$ (the correspondence with Shannon’s water-filling theorem). By the independence across dimensions, the relation between the optimal AU count and $d^*(\beta)$, as well as the monotone saturation, follows. $\square$

Remark 3. Implications: (i) raising $\beta\tau^2$ (stronger regularization or larger noise) reduces the number of active dimensions. (ii) A vanilla AE corresponds to the limit $\beta = 0$, where the threshold vanishes, so the number of utilized latent coordinates approaches m (consistent with experiment Exp-Syn-AU; Section 5.2). (iii) As a qualitative extension to real data, the theorem provides the rule of thumb that “dimensions whose effective SNR falls below β are compressed.”

Proposition 2 (Grid-Resolution Dependence of m^{knee} under the Idealized Monotone-Saturation Model). Assume the idealized model in which $AU(m)$ saturates ideally and monotonically at $d^*(\beta)$ (changing in a step-like manner, with no stochastic noise). Then, for $m^{\text{knee}}(\mathcal{G})$ on a grid $\mathcal{G} = \{m_1 < \dots < m_K\}$,

$$|m^{\text{knee}}(\mathcal{G}) - d^*(\beta)| \leq \max_i (m_{i+1} - m_i) \quad (6)$$

holds (the error is bounded above by the maximum grid spacing). 372

Caution: in practical non-linear VAEs this idealization does not hold, and shape noise in $AU(m)$ 373
due to stochastic optimization can produce errors that substantially exceed the upper bound in 374
Equation (6) (consistent with the observations in Section 5.3.4). This proposition should be treated 375
not as a general instability theorem but as a lemma providing guidance for grid design. 376

Proof. In the idealized monotone-saturation model, $AU(m)$ changes step-wise at $d^*(\beta)$, so 377
if $d^*(\beta) \in [m_{i_0}, m_{i_0+1})$, then m^{knee} is detected as m_{i_0+1} (or m_{i_0}). The error is $\leq m_{i_0+1} - m_{i_0} \leq$ 378
 $\max_i(m_{i+1} - m_i)$. $\square$ 379

Remark 4. Experimental implication of this proposition: the grid $\{4, 8, 12, 16, 20, 32, 64\}$ contains 380
a maximum spacing of 32 (from 32 to 64), so for $d^* \approx 10$ a large error with $m^{\text{knee}} \in \{32, 64\}$ 381
can arise (consistent with Section 5.3.4). A dense grid combined with a multi-seed ensemble is a 382
practical stabilization strategy (Section 6.2). 383

Proposition 3 (Pullback Metric and Local Isometry). A necessary and sufficient condition 384
for the decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$ to be locally isometric in a neighborhood of a point z is that the 385
pullback metric satisfies $G(z) = J_g(z)^\top J_g(z) = I_m$. That is, $\|J_g(z)dz\|^2 = \|dz\|^2$ holding for 386
every infinitesimal displacement dz is equivalent to $G(z) = I_m$ (all singular values equal to 1). 387

By contrast, the condition number $\kappa(z) = \sigma_{\max}(J_g) / \sigma_{\min}(J_g)$ is an indicator of the anisotropy 388
of the local map: $\kappa = 1$ means that all singular values are *equal* (isotropic scaling), but does not 389
guarantee that this scale equals 1. For example, if J_g magnifies all directions by a factor of 2, then 390
 $\kappa = 1$ but the map is not isometric. Accordingly, in this paper we use κ not as isometry itself but as 391
an auxiliary indicator of local anisotropy. 392

Proof. By the definition of isometry, $\|J_g dz\|^2 = \|dz\|^2$ for every $dz \in \mathbb{R}^m$ is equivalent to 393
 $J_g^\top J_g = I_m$ (all singular values equal to 1). Since $\kappa = 1$ only means that all singular values 394
are equal, whether their value is 1 is a separate matter; hence $\kappa = 1$ is not a necessary and 395
sufficient condition for isometry (see the example above). $\square$ 396

This proposition explains why an isometry bias arises in ID estimation using a refer- 397
ence AE (Section 6.7) and motivates the IsometricAE (experiment Exp-Syn-Iso). 398

The remaining interpretive propositions (effective rank of the decoder Jacobian, noise- 399
direction selectivity, and neighborhood-structure preservation) are given in A. 400

4.1. Qualitative Correspondence between Linear Theory and Non-Linear Experiments 401

Theorem 1 (the water-filling principle) holds under the linear-Gaussian rate-distortion 402
surrogate. In non-linear settings, a powerful decoder achieves an effective noise variance 403
 $\tau_{\text{eff}}^2 \ll 1$, so $d^*(\beta) = |\{j : \lambda_j > \beta \tau_{\text{eff}}^2\}|$ becomes substantially larger than the prediction 404
under the linear assumption ($\tau^2 = 1$). Moreover, the smooth AU growth curves produced 405
by non-linear optimization replace the step-like saturation transition of the linear setting, 406
causing the instability of m^{knee} on sparse grids. 407

We demonstrate this qualitative inheritance (AU monotonicity, slowdown of growth 408
beyond the effective dimensionality) and the quantitative divergence through a MNIST 409
PCA spectrum analysis (experiment Exp-Mnist-PCA) in Section 5.3.5. 410

4.2. Conditional Stability of TwoNN: Theoretical Formulation 411

In this section, motivated by the experimental observations of Section 5.2.4 (Exp-Stab, 412
Exp-Geom, and Exp-Iso-ID), we theoretically support, as an explanatory proposition, the 413
question “under what conditions can TwoNN be used as an auxiliary reference estimator?” 414
This proposition does not guarantee the unbiasedness of TwoNN; it is an interpretive 415
proposition giving a sketch of sufficient conditions for conditional stability (positioned 416

not as a mathematically rigorous theorem but as a theoretical rationale for a design guideline).

Proposition 4 (Conditional Stability of TwoNN). *Suppose the encoder $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$ is locally bi-Lipschitz, i.e., in a neighborhood U of every point z ,*

$$(1 - \varepsilon)\|v\| \leq \|J_f \cdot v\| \leq (1 + \varepsilon)\|v\|, \quad \varepsilon \in [0, 1] \quad (7)$$

holds, and suppose further that nearest-neighbor ranks are preserved with high probability (Trustworthiness $\geq 1 - \delta_T$ with small δ_T). Then the change in the empirical distribution of the two-nearest-neighbor distance ratio r_2/r_1 , measured by the earth mover's distance (EMD), is bounded as

$$\text{EMD}(P_{r_2/r_1}^{(\text{input})}, P_{r_2/r_1}^{(\text{latent})}) \leq C(\varepsilon, \delta_T) \quad (8)$$

and consequently the deviation of the TwoNN estimate, $|\hat{d}_{\text{ID}}^{(\text{latent})} - \hat{d}_{\text{ID}}^{(\text{input})}|$, is also bounded above by a constant governed by $C(\varepsilon, \delta_T)$. ($C(\varepsilon, \delta_T) \rightarrow 0$ as $\varepsilon \rightarrow 0$ and $\delta_T \rightarrow 0$.)

Proof. *Proof sketch:* By Equation (7), the change factor of distances $d(x_i, x_j)$ lies within the range $(1 - \varepsilon)^2$ to $(1 + \varepsilon)^2$. Since the distance ratio r_2/r_1 is a ratio rather than an absolute distance, uniform scale deformations (global rescaling) cancel out, but the non-isometric component with $\varepsilon \neq 0$ alters the ratio. Trustworthiness $\geq 1 - \delta_T$ guarantees that “the two nearest neighbors in latent space coincide with those in input space with high probability,” probabilistically suppressing large fluctuations of r_2/r_1 caused by rank reversals. Under these two conditions, the change in the distribution of r_2/r_1 is bounded above by a continuous function $C(\varepsilon, \delta_T)$ of ε and δ_T ($C = 0$ when $\varepsilon = 0$ and $\delta_T = 0$, coinciding with the isometric, perfectly neighbor-preserving case). (This is a finite-sample, finite-dimensional approximation; a rigorous bound depends on the specific setting of the isometry constants.) $\square$

Remark 5 (Experimental Implications of Proposition 4). *This proposition suggests that “isometry $\varepsilon \approx 0$ is the sole condition for TwoNN accuracy” is not the case; rather, “the improvement of Trustworthiness (preservation of neighbor ranks) induced by $m \gg d_{\text{ID}}$ may be the dominant factor for stability.” In experiment Exp-Stab (Section 5.2.4), the difference in TwoNN ID remained within $\pm 2.3\%$ over the range $\kappa \in [1.06, 2.16]$, whereas TwoNN recovered upon moving from $m = d_{\text{true}}$ to $m > d_{\text{true}}$ —consistent with the decrease in δ_T (the neighbor-reversal rate) being more dominant than the change in ε (isometry). This proposition does not claim that TwoNN is an “unbiased estimator of the true d_{ID} ”; it theoretically supports “under what conditions TwoNN can be operated as an auxiliary reference estimator.” Experiment Exp-Real-Geom (Section 5.10) directly validates this implication on real data.*

5. Experiments

Three-tier classification of experiments: The experiments in this paper are clearly classified into the following three tiers (corresponding to the “Category” column of Table 3).

Main experiments Empirical validation of the framework core: Exp-Syn-1, Exp-Syn-AU, Exp-Mnist-Base, Exp-Mnist-Multi, Exp-Mnist-Dense, Exp-Fashion, Exp-Real-Geom, and Exp-Nat-Cifar. Their primary validation targets are the robustness of AU values and TwoNN ID (RQ2) and the stabilization conditions of m^{knee} (RQ1).

Supplementary experiments Verification of prerequisites and interpretive claims of the main experiments: Exp-Stab, Exp-Geom, Exp-Iso-ID, Exp-Syn-Iso, Exp-Conv-AE, Exp-Syn-Top, Exp-Mnist-PCA, Exp-Mnist-Dyn, and Exp-Mnist-Ent. These verify

TwoNN numerical stability, IsometricAE isometry, the theory–experiment bridge, and related points (supporting evidence for the main claims).

Boundary experiments Quantification of the applicable range and boundary conditions: Exp-Conv-Fix, Exp-Conv-Ann, Exp-Conv-M256, Exp-Conv-M512, Exp-ConvV-HiBeta, Exp-NatImg-AU, Exp-NatImg-Sweep, and Exp-NatImg-Knee. These quantify the conditions under which stable identification of AU/m^{knee} becomes difficult for Conv-VAEs and natural images, thereby establishing the validity range and the applicability boundary of the framework.

5.1. Experimental Setup

Common settings: Adam ($\text{lr} = 10^{-3}$), batch size 128, ReLU activations (Sigmoid output layer), random seed 42, PyTorch 2.0.

Synthetic manifolds: hidden layers [64, 32], 200–300 epochs, VAE $\beta = 4.0$, $n = 5,000$ (4,000 training / 1,000 test).

MNIST / Fashion-MNIST: input dimension 784, hidden layers [512, 256], VAE $\beta = 4.0$, grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256\}$, batch size 128. MNIST is trained for 300 epochs ($n_{\text{train}} = 20,000$, $n_{\text{test}} = 3,000$), and Fashion-MNIST for 100 epochs ($n_{\text{train}} = 10,000$, $n_{\text{test}} = 2,000$).

ID estimation: TwoNN ($k = 2$; by the definition of the algorithm, $k = 2$ is the only possible choice) and MLE ($k = 10$; near the center of the range $k = 5$ –20 recommended by Levina & Bickel [9], with stability also confirmed for $k = 5, 10, 20$) [8,9]. AU threshold $\delta = 10^{-2}$ [21] (AU is extremely robust to this threshold choice: a sensitivity analysis confirmed that AU values remain essentially unchanged as δ varies over $[10^{-3}, 10^{-1}]$; Appendix G).

Computational environment and reproducibility: All experiments were run on a Linux workstation with a single NVIDIA GeForce RTX 3070 GPU (PyTorch 2.0, CUDA; falling back to CPU when CUDA is unavailable). The bulk of the computational cost is the repetition of training runs over the m grid $\times$ seeds; a major multi-seed sweep (e.g., MNIST FC-VAE with 7–17 grid points $\times$ 3 seeds $\times$ 300 epochs) takes several hours to about a day in this environment, and the full set of experiments amounts to several days of GPU time. Experiment code, configurations, and raw logs (JSON) are provided as supplementary material for review. A complete list of experimental configurations (dataset, model, epochs, β schedule, m grid, number of seeds, sample size, and role) is given in Table A10 of Appendix H, and **single-seed results are also marked as such in the captions of the corresponding tables and figures in the main text.**

Table 3. Summary of experiment groups. Detailed per-experiment configurations (epochs, β schedule, m grid, seeds, sample size, role) are listed in Table A10 (Appendix H). Categories: main experiments (core framework validation), supplementary experiments (prerequisite verification), boundary experiments (validity range quantification).

Experiment Group	Objective	Main Data	Category
Synthetic manifold experiments (Exp-Syn-1, Exp-Syn-AU, Exp-Syn-Iso, Exp-Syn-Top)	Principle verification (MSE elbow, AU saturation, isometry)	Swiss Roll, Torus, Möbius, etc.	Main + Supplementary
FC-AE/VAE experiments (Exp-Mnist-Base, Exp-Mnist-Multi, Exp-Mnist-Dense, Exp-Fashion, Exp-Stab, Exp-Geom, Exp-Iso-ID, Exp-Real-Geom)	Core validation of the multi-indicator diagnostic framework (robustness of AU values and TwoNN ID, m^{knee} stabilization, TwoNN applicability conditions)	MNIST, Fashion-MNIST, AE/VAE	Main
Conv-VAE experiments (Exp-Conv-AE, Exp-Conv-Fix, Exp-Conv-Ann, Exp-Conv-M256, Exp-Conv-M512, Exp-ConvV-HiBeta)	Identifying the cause of AU saturation delay in Conv-VAEs and determining the validity range	MNIST AE/VAE (Conv-	Supplementary + Boundary
Natural image experiments (Exp-Nat-Cifar, Exp-Nat-SVHN, Exp-NatImg-AU, Exp-NatImg-Sweep, Exp-NatImg-Knee)	Validation of external validity of TwoNN auxiliary estimates; quantification of applicability boundary in natural image settings	CIFAR-10, SVHN	Supplementary + Boundary

5.2. Synthetic Manifold Experiments: Principle Verification (RQ1, RQ2, RQ3)

Using synthetic manifolds with known d_{true} , we verify the fundamental validity of the proposed framework.

5.2.1. Correspondence Between MSE Elbows and Intrinsic Dimension (Experiment Exp-Syn-1)

For the Swiss Roll ($d_{\text{true}} = 2$, simply connected) and the Torus ($d_{\text{true}} = 2, d_{\text{emb}} = 3$), we confirmed that the AE reconstruction MSE improves sharply at $m = d_{\text{true}}$ (Swiss Roll) or $m = d_{\text{emb}}$ (Torus) (Table 4).

Table 4. Exp-Syn-1: MSE of AE on synthetic manifolds (mean±std over 3 seeds).

m	Swiss Roll MSE	Torus MSE
1	1.016 ± 0.315	0.356 ± 0.022
2	0.029 ± 0.001 ← elbow	0.109 ± 0.007
3	$(6 \pm 2) \times 10^{-4}$	$(4 \pm 3) \times 10^{-4}$ ← elbow
4	$(5 \pm 1) \times 10^{-4}$	$(1.0 \pm 0.4) \times 10^{-4}$

For the Swiss Roll, the MSE improves by a factor of approximately 35 from $m = 1$ to $m = 2$, experimentally confirming Proposition 1. For the Torus, the elbow appears at $m = 3$ ($= d_{\text{emb}}$), showing that perfect reconstruction is impossible at $d_{\text{ID}} = 2$.

5.2.2. AU Saturation and Topological Structure (Experiments Exp-Syn-AU and Exp-Syn-Top)

We compared the AU pattern of the VAE ($\beta = 4$) with that of a vanilla AE (Table 5). **Caveat:** the “AU” reported for the AE in the table is a pseudo-AU defined by whether the encoding variance of each latent coordinate exceeds δ (AU is not natively defined for AEs, which lack KL regularization); it is an auxiliary observation.

Table 5. Exp-Syn-AU: VAE active units (AU) for Swiss Roll (upper) and Torus (lower) (mean $\pm$ std, 3 seeds).

	m	Vanilla AE (AU)	VAE (AU, $\beta = 4$)
Swiss Roll ($d_{\text{true}} = 2$)	2	2	2.0 ± 0.0 (saturation)
	4	4	2.0 ± 0.0
	8	8	2.0 ± 0.0
	16	16	2.0 ± 0.0
Torus ($d_{\text{true}} = 2, d_{\text{emb}} = 3$)	4	4	2.0 ± 0.0
	6	6	3.0 ± 0.0 (saturation)
	8	8	3.0 ± 0.0
	16	16	3.0 ± 0.0

Whereas the vanilla AE exhibits $\text{AU} = m$ (no saturation), the VAE saturates at $\text{AU}_{\text{sat}} = 2$ ($= d_{\text{true}}$) for the Swiss Roll and at $\text{AU}_{\text{sat}} = 3$ ($= d_{\text{emb}} > d_{\text{ID}}$) for the Torus (standard deviation 0.0 across all 3 seeds). This is consistent with the qualitative implication of Theorem 1 (the water-filling AU saturation mechanism), and suggests that on the Torus the KL regularization may begin compressing near the topological embedding dimension (an empirical observation under the conditions of this study; note that this finding is preprocessing-dependent and conditional).

Table 6 shows the results of experiment Exp-Syn-Top (correspondence between AU_{sat} and d_{emb} on topological manifolds).

Table 6. Exp-Syn-Top: Topological manifolds and VAE AU_{sat} (standardized conditions; Swiss Roll excluded from the main evidence, see text).

Manifold	d_{ID}	π_1	d_{emb}	AU_{sat} (standardized)
Torus T^2	2	$\mathbb{Z}^2$	3	3 (main result)
Möbius Band	2	$\mathbb{Z}$	3	3 (main result)
Circle S^1	1	$\mathbb{Z}$	2	2 ($m \geq 6$, conditional)

Caveat: the AU_{sat} of the Swiss Roll is strongly preprocessing-dependent (2 under standardized conditions, 3 under other conditions), so we restrict the main evidence supporting the correspondence between AU_{sat} and d_{emb} to the Torus and the Möbius Band ($\text{AU}_{\text{sat}} = d_{\text{emb}} = 3$). Even this correspondence is an empirical observation that depends on our experimental conditions (standardized preprocessing, $\beta = 4$, sufficient convergence) and is not guaranteed as a theoretical necessity (see §6).

5.2.3. Achieving Isometry with IsometricAE (Experiment Exp-Syn-Iso)

On the Swiss Roll ($m = 2$), we compared the isometry of four model types (Table 7).

Table 7. Exp-Syn-Iso: Comparison of four models on Swiss Roll ($m = 2$) (mean $\pm$ std, 3 seeds).

Model	MSE	κ	NSR
AE	0.030 ± 0.003	1.251 ± 0.184	0.979 ± 0.036
CAE	0.036 ± 0.008	1.438 ± 0.105	0.659 ± 0.105
IsometricAE	0.029 ± 0.003	1.040 ± 0.009	0.753 ± 0.051
DAE	0.032 ± 0.003	1.179 ± 0.139	0.609 ± 0.085

Only IsometricAE achieves $\kappa \approx 1.04$ (isometry), consistent with Proposition 3, showing that an isometric latent space can be realized without substantially sacrificing MSE. An isometric latent space is a prerequisite that improves the reliability of ID estimation, and it contributes to the theoretical accuracy of Stage 1 ID estimation (with a trade-off against computational cost; §6.7).

5.2.4. Numerical Stability of TwoNN ID and the Effect of κ (Experiments Exp-Stab, Exp-Geom, and Exp-Iso-ID; details in Appendix E)

Three supplementary experiments—Exp-Stab (verifying the effect of κ on TwoNN estimation on the Swiss Roll), Exp-Geom (confirming TwoNN convergence via an MNIST AE m -sweep), and Exp-Iso-ID (a multi-dimension ID-estimation comparison between IsometricAE and a standard AE)—yielded a consistent message: **“the dominant factor for TwoNN stability is the condition $m \gg \hat{d}_{ID}$, not the magnitude of κ .”** Note that Exp-Stab and Exp-Iso-ID are separate experiments with independent data generation and training runs; their absolute TwoNN estimates (e.g., 1.55 vs. 1.78 for the AE at $m = 2$) are therefore not directly comparable, and only within-experiment relative comparisons are meaningful.

In Exp-Stab (Swiss Roll), at $m = 2 = d_{\text{true}}$ both the AE ($\kappa = 1.79$) and the IsometricAE ($\kappa = 1.06$) underestimate similarly with $\text{TwoNN} \approx 1.55$, whereas at $m = 4 > d_{\text{true}}$ TwoNN recovers to ≈ 2.03 (insufficient bottleneck margin is the primary cause; the effect of the κ difference is minor). In Exp-Geom (MNIST AE m -sweep), TwoNN converges stably to 10.16 for $m \geq 48 \approx 5\hat{d}_{ID}$, and even over $m = 64$ –128, where κ jumps from 4.25 to 37.0, TwoNN remains stable at 10.17–10.98 (detailed table in Appendix E, Table A2). In Exp-Iso-ID (Swiss Roll, $m \in \{2, 3, 4, 6, 8\}$), for $m \geq 3$ the differences in TwoNN ID remained within $\pm 2.3\%$ despite κ differences in $[1.1, 2.1]$ (detailed table in Appendix E, Table A3).

The findings of these three experiments reinforce the design rationale for using a reference AE ($m_{\text{ref}} = 256 \gg \hat{d}_{ID} \approx 10$) in Stage 1, and also show that there is a risk of underestimation for $m < 32 \approx 3\hat{d}_{ID}$ (note that “numerically stable” does not mean “geometrically accurate”; §6.7).

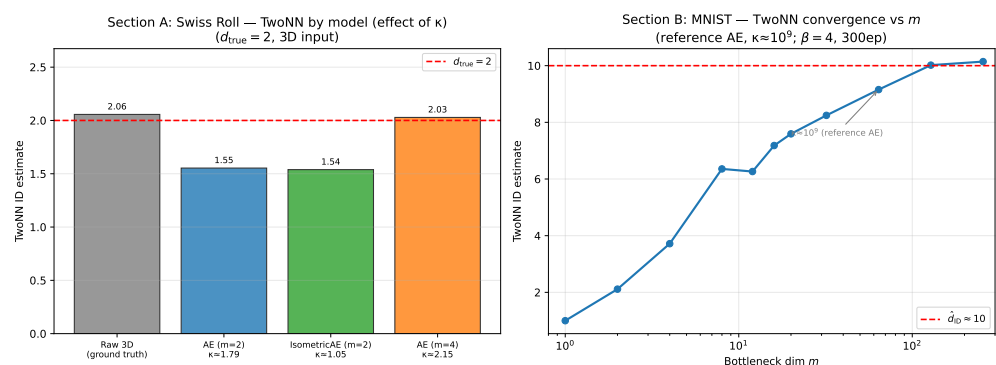


Figure 2. Exp-Stab/Exp-Geom: (Left) TwoNN ID on Swiss Roll — AE and IsometricAE both underestimate at $m = d_{\text{true}}$; TwoNN recovers at $m = 4 > d_{\text{true}}$. (Right) MNIST AE m -sweep: TwoNN stabilizes at $m \geq 48$ despite κ jumping to 37.0 at $m = 128$. Detailed tables in Appendix E.

5.3. MNIST: Reading Off the Effective Representation Dimensionality (Main Experiments, RQ1 and RQ2)

We validate the application of the proposed framework to real data using the MNIST dataset (main experiments). MNIST provides a realistic evaluation environment for the framework in the sense that it is “real data for which d_{true} is unknown but $\hat{d}_{ID}$ can be estimated.”

5.3.1. Intrinsic-Dimension Estimation with an AE (Experiment Exp-Mnist-Base-AE)

Table 8 shows the results for the MNIST AE (300 epochs).

Table 8. Exp-Mnist-Base: MNIST AE results (300 epochs, $n_{\text{train}} = 20,000$; **single seed 42**).

m	MSE	TwoNN ID	MLE ID	CKA
1	0.0437	1.02	1.15	0.102
4	0.0292	3.86	4.29	0.575
8	0.0191	6.08	6.70	0.725
16	0.0121	7.96	8.61	0.812
32	0.0087	8.84	9.89	0.875
64	0.0062	9.54	11.26	0.908
128	0.0061	9.84	11.45	0.921
256	0.0057	10.02	11.85	0.953

For $m \geq 64$, the TwoNN ID (9.54–10.02) and the MLE ID (11.26–11.85) converge, indicating that the effective intrinsic dimension of MNIST lies roughly in the range $\hat{d}_{\text{ID}} \approx 10$ –12.

Comparison with linear PCA: Applying PCA to MNIST ($n = 20,000$) requires 43 dimensions to reach 80% cumulative explained variance and 86 dimensions to reach 90%. These values are 4–8 times larger than the AE-based estimate (TwoNN ≈ 10), demonstrating the nonlinear manifold structure of MNIST (see Figure 3).

5.3.2. VAE AU Dynamics and m^{knee} (Experiment Exp-Mnist-Base-VAE)

Table 9 and Figure 3 show the AU pattern of the MNIST VAE ($\beta = 4$, 300 epochs).

Table 9. Exp-Mnist-Base: MNIST VAE ($\beta = 4$) results (300 epochs; **single seed 42**). AU increase rate slows at $m = 12$ ($m^{\text{knee}} = 12$; preliminary, not reproduced in multi-seed runs).

m	MSE	AU	TwoNN ID	Behavior of $\rho(m)$
1	0.0464	1	1.00	—
4	0.0275	4	3.72	normal growth
8	0.0186	8	6.36	normal growth
12	0.0184	8	6.26	sharp drop $\leftarrow m^{\text{knee}} = 12$
16	0.0148	11	7.18	low
20	0.0134	13	7.59	low
32	0.0121	15	8.25	low
64	0.0103	29	9.16	gradual
128	0.0085	26	10.02	gradual
256	0.0072	36	10.14	gradual

At $m = 12$, $\text{AU} = 8$ (the same value as at the preceding $m = 8$), and the normalized increase rate $\rho(m)$ drops sharply ($\rho(12) \approx 0$). This triggers the threshold $\tau_{\text{knee}} = 0.25$, and $m^{\text{knee}} = 12$ is observed.

Caveat: this $m^{\text{knee}} = 12$ is a preliminary observation obtained with a single seed (seed 42) and a specific extended grid ($m \in \{1, 4, 8, \dots, 256\}$), and it does not reproduce in the multi-seed experiments (Section 5.3.4) (the main stabilized result of this paper is Exp-Mnist-Dense-300; Section 5.3.6).

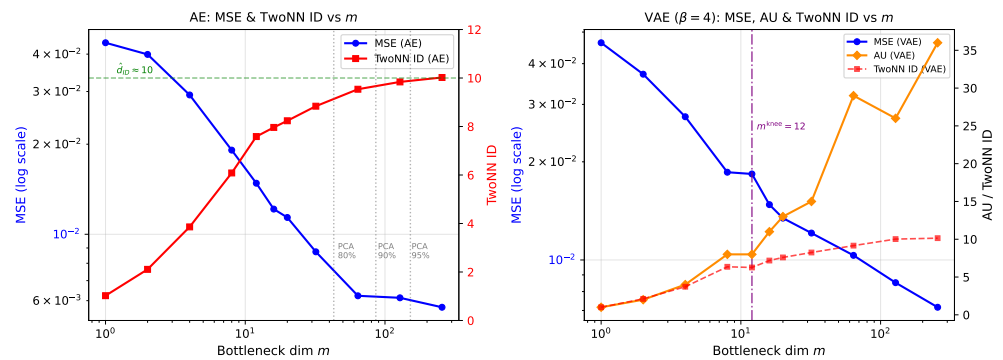


Figure 3. Exp-Mnist-Base: m -dependence of each indicator for MNIST (300 epochs). **Left (AE):** MSE (log scale, left axis) and TwoNN ID (right axis). Vertical dotted lines correspond to PCA cumulative variance 80% (43 dims), 90% (86 dims), 95% (152 dims); the difference from the nonlinear AE estimate (TwoNN ≈ 10) is clear. **Right (VAE, $\beta = 4$):** MSE (log scale, left axis), AU (right axis), TwoNN ID (right axis, dashed). At $m = 12$ (m^{knee}), the AU increase rate drops sharply (arrow), indicating a heuristic slowdown point; the main stabilized result is $m^{\text{knee}} = 10.3 \pm 0.9$ from Exp-Mnist-Dense-300 (Section 5.3.6).

5.3.3. Comparison with Existing Indicators

We compare the proposed indicator m^{knee} with other representative indicators for choosing the bottleneck dimension (Table 10).

Table 10. Comparison of bottleneck dimension estimates on MNIST. m^{knee} entries distinguish sparse-grid and dense-grid conditions.

Method	Estimate	Source	Characteristics / Limitations
PCA (80% var.)	43	Exp-Mnist-Base-AE	Linear; overestimates nonlinear ID
PCA (90% var.)	86	Exp-Mnist-Base-AE	Linear; larger overestimate
MSE elbow (AE)	≈ 16 –20	Exp-Mnist-Base-AE	Intuitive; depends on smoothing
TwoNN ID (ref. AE)	≈ 10 –12	Exp-Mnist-Base-AE	Nonlinear; subject to isometry bias
m^{knee} sparse-grid (3 seeds)	43 ± 15	Exp-Mnist-Multi-300	Exploratory; unreliable at sparse grid
m^{knee} dense-grid (3 seeds)	10.3 ± 0.9	Exp-Mnist-Dense-300	TwoNN-guided local refinement; requires dense grid + multi-seed

Dense-grid: $m \in \{8, 9, \dots, 16, 18, \dots, 64\}$ with independent initialization per (m, seed) ; the dense region was placed around the TwoNN estimate (≈ 10), so the agreement is a TwoNN-guided refinement, not an independent validation. Sparse-grid $m^{\text{knee}} = 43 \pm 15$ is unreliable as a diagnostic reference.

PCA, being linear, cannot capture the nonlinear manifold structure and requires more than four times as many dimensions as the AE-based estimate, while the MSE elbow tends to be ambiguous on smooth MSE curves. m^{knee} is complementary to these indicators in that it directly observes the KL dynamics, but it depends on τ , β , and the m -grid, and its use presupposes the multi-indicator approach.

5.3.4. Multi-Seed Validation (Experiments Exp-Mnist-Multi-150 and Exp-Mnist-Multi-300)

To assess the statistical reliability of m^{knee} and the influence of the degree of convergence, we trained the MNIST VAE ($\beta = 4$, $m \in \{4, 8, 12, 16, 20, 32, 64\}$) with three random seeds (42, 123, 777) under two conditions: 150 epochs (Exp-Mnist-Multi-150) and 300 epochs (Exp-Mnist-Multi-300). Table 11 shows the 300-epoch results (the 150-epoch m^{knee} values are summarized in Table 12; the AU and TwoNN ID trends are identical to the 300-epoch condition, with the TwoNN ID at $m = 64$ being 9.94 ± 0.08).

Table 11. Exp-Mnist-Multi-300: Mean $\pm$ std of AU and TwoNN ID for 3 seeds at 300 epochs ($\beta = 4$). AU and TwoNN ID are highly consistent, but m^{knee} still varies.

m	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	Note
4	4.0 ± 0.0	3.98 ± 0.14	all dims active
8	8.0 ± 0.0	6.67 ± 0.15	
12	10.0 ± 0.0	7.41 ± 0.06	
16	11.7 ± 0.5	7.95 ± 0.21	AU still increasing
20	13.3 ± 0.5	8.37 ± 0.14	
32	15.7 ± 0.5	9.07 ± 0.10	
64	22.3 ± 0.5	10.03 ± 0.12	converges to $\hat{d}_{\text{ID}} \approx 10$
$m^{\text{knee}} (\tau = 0.25)$: seed 42 = 64, seed 123 = 32, seed 777 = 32			
Mean $m^{\text{knee}} = 43 \pm 15$ (300 epochs)			

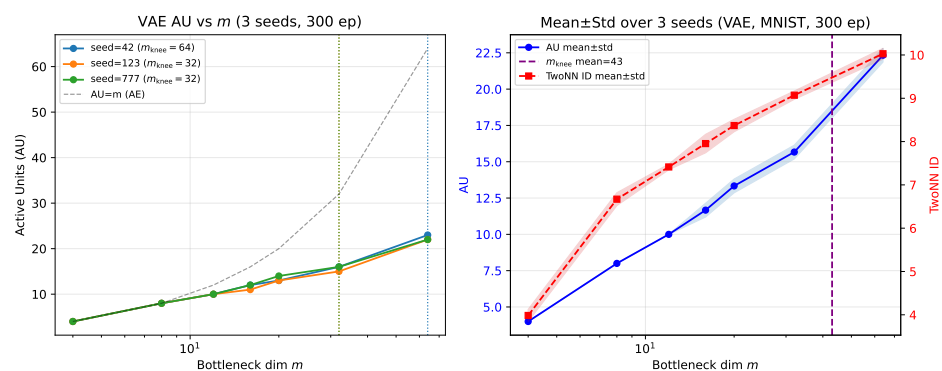


Figure 4. Exp-Mnist-Multi: AU patterns of MNIST VAE (3 seeds, 300 epochs). Left: AU vs m per seed (dotted lines: m^{knee}); Right: mean $\pm$ std band. AU and TwoNN ID bands are very narrow, but m^{knee} varies between 32 and 64 across seeds.

Key findings: The AU values (Table 11) are highly consistent across the three seeds under both epoch conditions ($\sigma \leq 0.5$), and the TwoNN ID is likewise stable at 10.03 ± 0.12 for $m = 64$ (300 epochs). In contrast, m^{knee} ($\tau = 0.25$) varied over the range 32–64 in both conditions: 53 ± 15 at 150 epochs (per-seed 64, 32, 64) and 43 ± 15 even under full 300-epoch convergence (per-seed 64, 32, 32). This occurs because the normalized increase rate $\rho(m)$ fluctuates subtly around the threshold $\tau = 0.25$ over several values of m , and extending training does not resolve it—the **dominant cause of the m^{knee} instability is not insufficient convergence but the dependence on the m -grid and the random seed**. The $m^{\text{knee}} = 12$ observed in the extended Exp-Mnist-Base experiment (single seed 42) also fails to reproduce under these conditions and remains a preliminary observation dependent on the grid design and initialization trajectory. In this paper we treat $m^{\text{knee}} = 10.3 \pm 0.9$ from Exp-Mnist-Dense-300—obtained with a dense grid around the TwoNN estimate, independent initialization, and multiple seeds—as the main stabilization result (Section 5.3.6).

5.3.5. Linear Theory–Experiment Bridge (Experiment Exp-Mnist-PCA; Details in Appendix E)

Contrasting the MNIST PCA spectrum ($\lambda_1 = 5.22, \dots, \lambda_{10} = 1.23$) with Theorem 1, we obtain $d^* \approx 10$ for $\tau_{\text{eff}}^2 \approx 0.30$ (back-calculated), qualitatively consistent with the observed TwoNN ID. Whereas the linear setting produces a sharp saturation transition, in nonlinear VAEs gradient competition yields a smooth, gradually increasing curve, which theoretically corroborates the sparse-grid instability of m^{knee} (detailed PCA spectrum tables and theory–experiment comparison tables are given in Appendix E). For Conv-VAEs, the powerful reconstruction capability of the nonlinear decoder makes τ_{eff}^2 even smaller, so the m required for AU saturation increases substantially relative to the FC-VAE (Section 6.6).

5.3.6. TwoNN-Guided Local Refinement with a Dense Grid (Experiment Exp-Mnist-Dense)

To identify which factors cause the instability of m^{knee} , we designed an experiment that isolates three factors: grid density, epochs, and seed. Exp-Mnist-Base and Exp-Mnist-Multi already separate the epoch factor (150 ep vs. 300 ep) and the seed factor (3 seeds). The present experiment, Exp-Mnist-Dense, further isolates the grid-density factor: we trained with $m \in \{8, 9, 10, 11, 12, 13, 14, 15, 16, 18, 20, 24, 28, 32, 40, 48, 64\}$ (17 points, adjacent spacing 1–16; step 1 over $m = 8$ –16 near the estimated ID) for 300 epochs and 3 seeds. **Explicit positioning:** the dense region ($m = 8$ –16) was placed around the Stage-1 TwoNN estimate ($\hat{d}_{\text{ID}} \approx 10$), so the choice of the candidate region itself depends on the TwoNN estimate of the same framework. Consequently, the agreement between m^{knee} and the TwoNN ID reported below demonstrates the success of a **TwoNN-guided local refinement**, not a mutual validation independent of TwoNN (securing independence—via a pre-registered dense grid over a broader range, or nested validation in which the search region is estimated and m^{knee} is evaluated on disjoint seeds or data subsets—is left as future work).

Table 12. Exp-Mnist-Dense: Comparison of m^{knee} between dense and sparse grids ($\tau = 0.25$, MNIST FC-VAE, $\beta = 4$). Consistent with the grid-sensitivity bound of Proposition 2.

Condition	Max. grid interval	seed=42	seed=123	seed=777	Mean $\pm$ std
Sparse grid, 150ep (Exp-Mnist-Multi-150)	32	64	32	64	53 ± 15
Sparse grid, 300ep (Exp-Mnist-Multi-300)	32	64	32	32	43 ± 15
Dense, 300ep (Exp-Mnist-Dense-300)	16 (step 1, 8–16)	11	9	11	10.3 ± 0.9

Table 13. Exp-Mnist-Dense: Per-seed AU(m) curves for the full dense grid (300ep, all 3 seeds). The AU curves fluctuate non-monotonically in the transition region ($m = 11$ –16).

m	8	9	10	11	12	13	14	15	16	18	20	24	28	32	40	48	64
AU (seed 42)	8	9	10	10	10	10	12	11	14	13	15	14	15	15	18	19	21
AU (seed 123)	8	8	10	9	11	10	11	13	12	14	13	15	14	17	19	19	21
AU (seed 777)	8	9	10	10	11	10	11	11	12	12	13	15	16	18	18	20	21

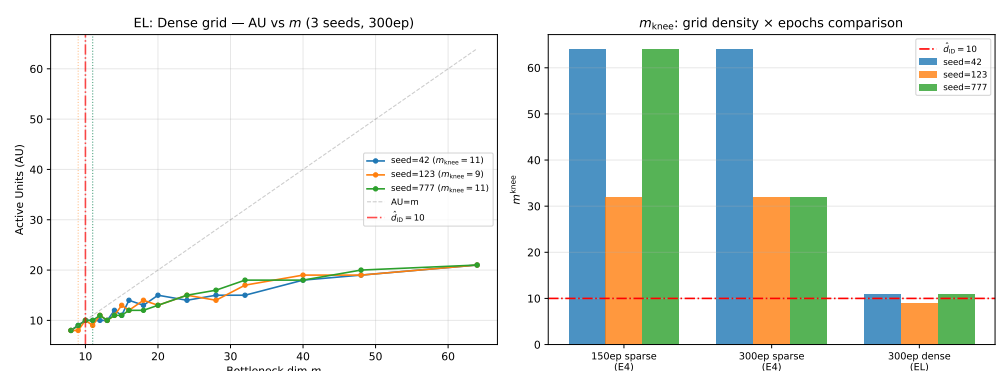


Figure 5. Exp-Mnist-Dense: (Left) Dense-grid AU vs m (3 seeds). (Right) Comparison by grid density $\times$ epochs. m^{knee} variation in the dense grid can be assessed by the upper bound of Proposition 2 (maximum grid interval).

Significance of the Exp-Mnist-Dense experiment: The three-row comparison in Table 12 quantitatively isolates the three factors (grid density, epochs, and seed):

- **Epoch factor:** sparse grid, 150 ep $\rightarrow$ 300 ep (m^{knee} : $53 \pm 15 \rightarrow 43 \pm 15$) — increasing the epochs stabilizes m^{knee} slightly, but is not the main cause

- **Seed factor:** variation across the 3 seeds under identical conditions ($\sigma = 15$) — differences in optimization trajectories due to initialization matter, but this is a secondary factor that coexists with the sparse grid
- **Grid-density factor (one of the main causes):** with the dense grid (Exp-Mnist-Dense), $m^{\text{knee}} = 10.3 \pm 0.9$ (per-seed values 11, 9, 11) — a large improvement over the sparse-grid 43 ± 15 , consistent with the estimate $\hat{d}_{\text{ID}} \approx 10$ (note, however, that this is agreement as a TwoNN-guided refinement, since the dense region was placed around the TwoNN estimate)

Effect of grid density (comparison with the implication of Proposition 2): Proposition 2 is an auxiliary upper bound for an idealized monotone saturation model and does not apply directly to the empirical AU curves of nonlinear VAEs. Nevertheless, densely probing the neighborhood of the estimated ID ($m = 8\text{--}16$) in steps of 1 yielded $m^{\text{knee}} \in \{9, 11\}$ (3 seeds), greatly reducing the deviation from the TwoNN ID (≈ 10). This is qualitatively consistent with the implication of Proposition 2 — demonstrated for the idealized model — that grid design matters. However, since empirical AU curves exhibit non-monotone fluctuations (e.g., $\text{AU} = 10 \rightarrow 10 \rightarrow 10 \rightarrow 12 \rightarrow 11$ in Table 13), it is more accurate to state this as the observation that “densifying the grid greatly reduced the variance of the empirical m^{knee} ” rather than to claim that the deviation “stayed within the upper bound.”

Comparison of the τ -method and the second-difference method: Besides the first-difference ratio method adopted in this paper ($\tau = 0.25$), m^{knee} can be detected by the **second-difference method**, which locates the point of maximum curvature of the $\text{AU}(m)$ curve (Appendix C). Table 14 shows the comparison on the Exp-Mnist-Dense dense-grid data.

Table 14. Exp-Mnist-Dense: Comparison of τ -method ($\tau = 0.25$) and second-difference method (d2) for m^{knee} detection (dense grid, 300ep, 3 seeds).

Method	seed=42	seed=123	seed=777	Mean $\pm$ std
τ -method ($\tau = 0.25$)	11	9	11	10.3 ± 0.9
Second-difference method (d2)	15	28	12	18.3 ± 6.8
Reference: $\hat{d}_{\text{ID}} \approx 10$ (TwoNN, $m = 64$, 3-seed mean)				

The τ -method yields a mean of 10.3 ± 0.9 , consistent with $\hat{d}_{\text{ID}} \approx 10$, with small seed-to-seed variation. In contrast, the d2 method is sensitive to the discrete noise of $\text{AU}(m)$; for seed=123 it deviated strongly to $m^{\text{knee}} = 28$ (in the transition region the AU curve fluctuates as $8 \rightarrow 10 \rightarrow 9 \rightarrow 11 \rightarrow 10 \rightarrow 11 \rightarrow 13$, which perturbs the location of the maximum second difference). Therefore, **the τ -method is the more stable choice for detecting m^{knee} under dense-grid conditions**, and we position the d2 method as an auxiliary tool for locating global “elbow” candidates (see Appendix C).

Finally, the fact that the MNIST VAE reaches $\text{AU}_{\text{sat}} = 36$ ($m = 256$), substantially exceeding $\hat{d}_{\text{ID}} \approx 10\text{--}12$ (the Manifold Entanglement hypothesis), and the two-phase pattern of the training dynamics (Exp-Mnist-Dyn) are both single-seed, preliminary observations; multi-seed validation of these is left for future work (see Section 6.5).

5.4. Application to Fashion-MNIST (Experiment Exp-Fashion, Preliminary External Check)

Fashion-MNIST (10 clothing classes, 28×28) is positioned as a preliminary external check of the framework’s seed-reproducibility claims (the main experiments use MNIST with 3 seeds). It is a single-seed (seed 42) confirmation, and we do not claim the same level of reproducibility as for MNIST (3 seeds).

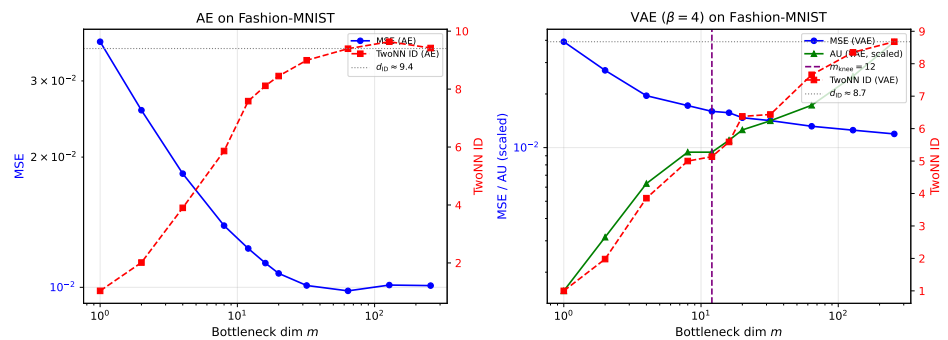


Figure 6. Exp-Fashion: m -dependence of MSE, AU, and TwoNN ID for Fashion-MNIST AE (left) and VAE (right) (100 epochs). In the VAE, the AU growth rate drops sharply around $m = 12$ (m^{knee} at $\tau_{\text{knee}} = 0.25$), and TwoNN ID asymptotically converges to ≈ 9.4 (AE) and ≈ 8.7 (VAE) at $m \geq 32$.

Table 15. Exp-Fashion: Fashion-MNIST AE and VAE results (100 epochs, $n_{\text{train}} = 10,000$, representative m values; **single seed 42**).

AE			VAE ($\beta = 4$)		
m	MSE	TwoNN ID	m	MSE	AU
4	0.01831	3.90	4	0.01955	4
8	0.01389	5.86	8	0.01721	6
12	0.01230	7.58	12	0.01597	6
16	0.01138	8.11	16	0.01570	7
20	0.01077	8.45	20	0.01471	8
32	0.01010	8.99	32	0.01418	9
64	0.00981	9.39	64	0.01318	11
256	0.01009	9.42	256	0.01193	25

The results of Experiment Exp-Fashion (Table 15, Figure 6) yield the following findings. For the AE, pseudo-AU = m (essentially all latent coordinates are active) and TwoNN ID converges to ≈ 9.4 at $m \geq 32$, suggesting $\hat{d}_{\text{ID}} \approx 9$ –10 for Fashion-MNIST. For the VAE ($\beta = 4$, single seed), $m^{\text{knee}} = 12$ ($\tau_{\text{knee}} = 0.25$) was observed; however, this is an exploratory single-seed observation, and we note that it does not carry the same level of reproducibility as the multi-seed sparse-grid result on MNIST ($m^{\text{knee}} = 43 \pm 15$). AU remains at 25 even at $m = 256$ (unlike 36 for MNIST), and the TwoNN ID estimates (≈ 9 –10) are also somewhat smaller than those for MNIST (≈ 10 –12). Interpreting these differences (e.g., in terms of dataset-level local degrees of freedom) would require a more rigorous analysis; here we report them only as single-seed observations.

5.5. Extension to Convolutional Architectures (Experiment Exp-Conv-AE)

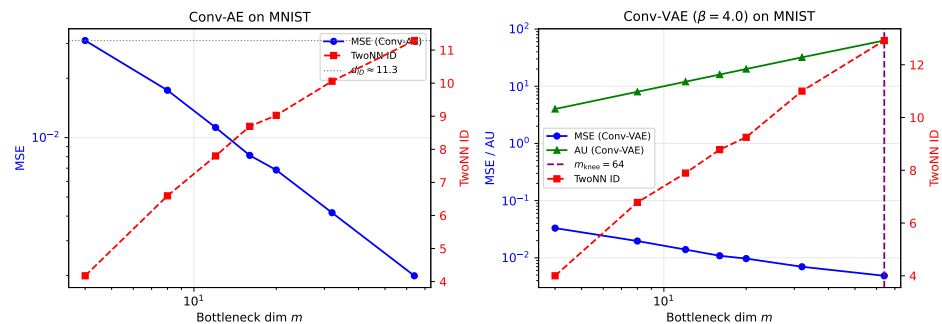
To address concerns about generalizability when the study is limited to fully connected architectures, we conducted an experiment applying convolutional AE/VAE (Conv-AE/VAE) to MNIST (Experiment Exp-Conv-AE).

Conv-AE/VAE architecture:

- Encoder: Conv($1 \rightarrow 32$, $k = 3$, stride= 2) $\rightarrow$ Conv($32 \rightarrow 64$, $k = 3$, stride= 2) $\rightarrow$ FC($3136 \rightarrow m$) (for the VAE, $\rightarrow \mu, \log \sigma^2$)
- Decoder: FC($m \rightarrow 3136$) $\rightarrow$ ConvT($64 \rightarrow 32$, $k = 4$, stride= 2) $\rightarrow$ ConvT($32 \rightarrow 1$, $k = 4$, stride= 2) $\rightarrow$ Sigmoid
- $n_{\text{train}} = 20,000$, 100 epochs, $m \in \{4, 8, 12, 16, 20, 32, 64\}$, seed 42

Table 16. Exp-Conv-AE: MNIST Conv-AE and Conv-VAE ($\beta = 4$) results (100 epochs, seed=42).

m	Conv-AE		m	Conv-VAE ($\beta = 4$)	
	MSE	TwoNN ID		MSE	AU
4	0.0311	4.18	4	0.0331	4
8	0.0174	6.59	8	0.0197	8
12	0.0113	7.80	12	0.0140	12
16	0.0081	8.69	16	0.0109	16
20	0.0069	9.03	20	0.0097	20
32	0.0042	10.06	32	0.0070	32
64	0.0020	11.29	64	0.0049	63

**Figure 7.** Exp-Conv-AE: m -dependence of MSE, AU, and TwoNN ID for MNIST Conv-AE (left) and Conv-VAE (right) (100 epochs). Conv-AE confirms $AU = m$ (all dims active) with TwoNN ID converging to ≈ 11 at $m \geq 32$. Conv-VAE shows $AU \approx m$ (no saturation due to insufficient convergence at 100 epochs).

Results of Experiment Exp-Conv-AE (Table 16): the **Conv-AE TwoNN ID** converges to 11.29 at $m = 64$, consistent with the FC-AE estimate (≈ 10 , within $\approx 10\%$)—TwoNN estimates are consistent across architectures (an empirical observation under the conditions of this study). The **Conv-VAE**, in contrast, shows $AU = m$ (no saturation) for $m \leq 32$: at 100 epochs the KL-driven dimension suppression is not yet functioning, and reliable detection of m^{knee} requires a larger number of epochs.

5.5.1. Delayed AU Saturation in Conv-VAEs: Fixed- β vs. KL-Annealing Comparison (Experiments Exp-Conv-Fix and Exp-Conv-Ann)

Following the insufficient convergence in Exp-Conv-AE, we verified with 300 epochs and 3 seeds (Exp-Conv-Fix) and with KL cyclic annealing (Exp-Conv-Ann) (Table 17).

Table 17. Exp-Conv-Fix and Exp-Conv-Ann: Comparison of AU for MNIST Conv-VAE (300ep, mean $\pm$ std over 3 seeds). FC-VAE (Exp-Mnist-Multi-300) slows down by $m = 32$ ($AU = 15.7$), while Conv-VAE shows $AU = m$ (no saturation) under both fixed- β and annealing at $m \leq 32$.

m	Conv-VAE fixed- β (Exp-Conv-Fix)		Conv-VAE annealing (Exp-Conv-Ann)		FC-VAE (Exp-Mnist-Multi-300)	
	AU	TwoNN ID	AU	TwoNN ID	AU	TwoNN ID
4	4.0 ± 0.0	3.97 ± 0.08	4.0 ± 0.0		4.0	3.98
12	12.0 ± 0.0	8.14 ± 0.09	12.0 ± 0.0		10.0	7.41
32	32.0 ± 0.0	11.43 ± 0.08	32.0 ± 0.0		15.7	9.07
64	54.3 ± 0.5	13.10 ± 0.05	64.0 ± 0.0		22.3	10.03
m^{knee} : Conv-VAE (both conditions) all seeds = 64 (grid upper bound); FC-VAE = 43 ± 15 (32–64)						

The Conv-VAE exhibits $AU = m$ (no saturation) for $m \leq 32$ under both fixed $\beta = 4$ and cyclic annealing, making the divergence from the FC-VAE clear (theoretical discussion in §6.6). $AU \approx m$ is a signal that “KL regularization is not yet functioning,” and in this state

the TwoNN ID is also inflated (Conv-VAE $m = 64$: 13.10 vs. FC-VAE: 10.03). **On the grid**
 $m \leq 64$, **AU saturation cannot be observed**, and an extension of the m range is required.

5.6. Conv-VAE Extended m -Grid Experiments (Experiments Exp-Conv-M256 and Exp-Conv-M512)

Since no AU saturation was observed for $m \leq 64$ in Exp-Conv-Fix and Exp-Conv-Ann (§5.5.1), we extended the m grid to 256 (Experiment Exp-Conv-M256) and 512 (Experiment Exp-Conv-M512). Under the reduced- τ_{eff}^2 hypothesis discussed in §6.6, the Conv-VAE's $d^*(\beta)$ is substantially larger than that of the FC-VAE (≈ 10), so a slowdown of AU growth is predicted to appear in the region $m > 64$.

Experimental setup (Exp-Conv-M256): Using the same Conv-VAE architecture as Exp-Conv-Ann (28×28 MNIST, 2-layer Conv encoder, 2-layer ConvTranspose decoder, KL cyclic annealing with $\beta_{\text{max}} = 4$, 3 cycles, 150 epochs, seeds $\in \{42, 123, 777\}$), we tested $m \in \{4, 8, 16, 32, 64, 96, 128, 192, 256\}$ (9 points).

Table 18. Exp-Conv-M256: Conv-VAE extended m grid (KL annealing, 150 epochs, mean $\pm$ std, 3 seeds). AU = m for $m \leq 64$; AU < m begins at $m \geq 96$. FC-VAE reference (Exp-Mnist-Multi-300, $m = 64$): AU = 22.3 ± 0.5 .

m	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	Note
4	4.0 $\pm$ 0.0	3.84 $\pm$ 0.11	all dims active
8	8.0 $\pm$ 0.0	6.64 $\pm$ 0.11	
16	16.0 $\pm$ 0.0	8.76 $\pm$ 0.05	
32	32.0 $\pm$ 0.0	10.46 $\pm$ 0.13	
64	64.0 $\pm$ 0.0	13.29 $\pm$ 0.12	
96	90.7 $\pm$ 1.2	14.30 $\pm$ 0.22	same as Exp-Conv-Ann (no saturation) AU < m onset (FC-VAE $\hat{d}_{\text{ID}} \approx 9 \times$)
128	111.7 $\pm$ 2.6	14.69 $\pm$ 0.13	
192	159.0 $\pm$ 4.3	15.89 $\pm$ 0.27	
256	241.3 $\pm$ 2.9	16.58 $\pm$ 0.16	

m^{knee} ($\tau = 0.25$): all seeds = 256 (no knee within the tested range)

FC-VAE reference (Exp-Mnist-Multi-300, $m = 64$): AU = 22.3 ± 0.5 , TwoNN ID = 10.03 ± 0.12

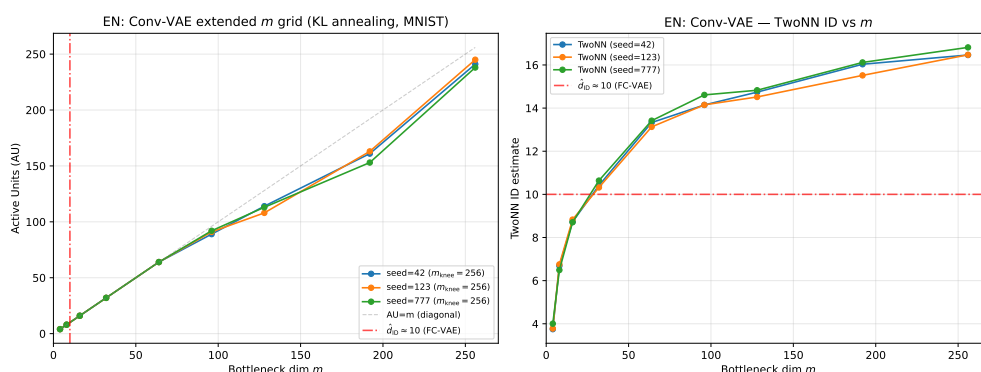


Figure 8. Exp-Conv-M256: Conv-VAE extended m grid ($m \in [4, 256]$, KL annealing, 3 seeds). AU vs m (left) and TwoNN ID vs m (right). Diagonal line (AU = m) and FC-VAE reference $\hat{d}_{\text{ID}} \approx 10$ (red) are shown. AU = m continues at $m \leq 64$ but deviates from the diagonal at $m \geq 96$, confirming qualitative water-filling behavior in Conv-VAE.

Main findings of Experiment Exp-Conv-M256: For $m \leq 64$, AU = m holds across all 3 seeds (in stark contrast to the FC-VAE's AU = 22.3 at $m = 64$); a saturation trend first appears at $m = 96$ (AU = $90.7 \pm 1.2 < m$), but even at $m = 256$ the activation rate remains high (94.3%) and m^{knee} stays pinned to the grid boundary (256) for all seeds. This indicates that the Conv-VAE's $d^*(\beta)$ is substantially larger than the FC-VAE's (≈ 10), exceeding 256 (the correspondence with the τ_{eff}^2 theory is discussed in §6.6).

Extension beyond $m = 256$ (Experiment Exp-Conv-M512, details in Appendix F): Even when extending to $m \in \{384, 512\}$, $AU \approx m$ (all dimensions active) is maintained across all 3 seeds, and m^{knee} remains pinned to the grid boundary at 512 (see Appendix F, Table A6). This shows that the $d^*(\beta)$ of the MNIST Conv-VAE exceeds $m = 512$, consistent with the τ_{eff}^2 theoretical prediction of §6.6. In the next subsection, we show that AU saturation can be artificially induced by strengthening β_{max} .

5.7. Artificially Inducing Conv-VAE AU Saturation with High β (Experiment Exp-ConvV-HiBeta)

As a verification of the τ_{eff}^2 hypothesis of §6.6 (a powerful decoder pushes AU saturation toward higher dimensions), we increased β_{max} from 4 to $\{10, 20\}$ and checked whether an AU saturation plateau is induced in the MNIST Conv-VAE.

Experimental setup: Using the same architecture and KL cyclic annealing (3 cycles) as Exp-Conv-M256, we trained with $\beta_{\text{max}} \in \{10, 20\}$, $m \in \{16, 32, 64, 96, 128, 192, 256\}$, seed=42, and 150 epochs (the $\beta_{\text{max}} = 4$, seed=42 results of Exp-Conv-M256 are reproduced for comparison).

Table 19. Exp-ConvV-HiBeta: Comparison of MNIST Conv-VAE AU at $\beta_{\text{max}} \in \{4, 10, 20\}$ (seed=42, 150ep, KL cyclic annealing). Higher β_{max} induces AU saturation at lower dimensions.

m	$\beta_{\text{max}} = 4$ (Exp-Conv-M256)	$\beta_{\text{max}} = 10$	$\beta_{\text{max}} = 20$
16	16 (100%)	16 (100%)	16 (100%)
32	32 (100%)	32 (100%)	27 (84%)
64	64 (100%)	54 (84%)	42 (66%)
96	89 (93%)	65 (68%)	47 (49%)
128	114 (89%)	69 (54%)	53 (41%)
192	161 (84%)	79 (41%)	60 (31%)
256	241 (94%)	93 (36%)	66 (26%)
$m^{\text{knee}} (\tau = 0.25)$	512 (grid boundary, unidentified)	≈ 128	≈ 96

Findings of Exp-ConvV-HiBeta (Table 19): Whereas at $\beta_{\text{max}} = 4$ $AU = m$ (100% active) persists for $m \leq 64$, at $\beta_{\text{max}} = 10$ a clear AU saturation trend appears — $AU = 54$ (84%) at $m = 64$, $AU = 69$ (54%) at $m = 128$, and $AU = 93$ (36%) at $m = 256$ — and $m^{\text{knee}} \approx 128$ ($\tau = 0.25$) was identified. Furthermore, at $\beta_{\text{max}} = 20$, saturation occurs at even lower dimensions — $AU = 27$ (84%) at $m = 32$ and $AU = 42$ (66%) at $m = 64$ — and $m^{\text{knee}} \approx 96$ was observed. This directly matches the prediction of the τ_{eff}^2 hypothesis of §6.6: increasing β shifts $d^*(\beta) = |\{j : \lambda_j > \beta \tau_{\text{eff}}^2\}|$ to smaller values, and AU saturation appears at lower dimensions. **This result is consistent with the interpretation that the absence of observed AU saturation in the MNIST Conv-VAE is not an “applicability limit of the framework,” but rather stems from the fact that at the setting $\beta_{\text{max}} = 4$ the regularization was insufficient relative to the decoder capacity, resulting in $d^*(\beta) \gg 256$.** Applying the AU/m^{knee} framework to Conv-VAEs requires an appropriate setting of β matched to the data and architecture.

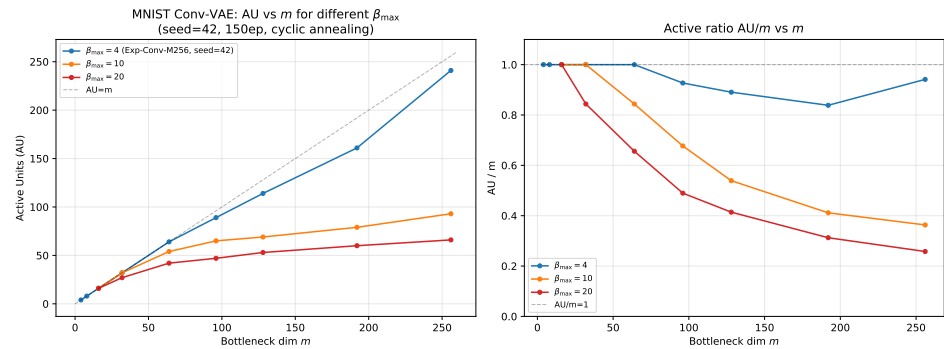


Figure 9. Exp-ConvV-HiBeta: Comparison of MNIST Conv-VAE AU vs m (left) and activation rate AU/ m (right) for $\beta_{\max} \in \{4, 10, 20\}$. Increasing $\beta_{\max}$ shifts AU saturation to lower dimensions, demonstrating the β -dependence of $d^*(\beta)$.

5.8. Application to Natural Images: CIFAR-10 and SVHN (Experiments Exp-Nat-Cifar and Exp-Nat-SVHN)

We applied the proposed framework to the more complex natural-image datasets CIFAR-10 [3] and SVHN [4], and compared the results with the intrinsic-dimension estimates reported by Pope et al. [16].

Experimental setup (Exp-Nat-Cifar: CIFAR-10; Exp-Nat-SVHN: SVHN):

- Architecture: Conv-VAE (input $32 \times 32 \times 3$)
 - Encoder: Conv($3 \rightarrow 32, k=4, s=2, p=1$) $\rightarrow$ Conv($32 \rightarrow 64, k=4, s=2, p=1$) $\rightarrow$ Conv($64 \rightarrow 128, k=4, s=2, p=1$) $\rightarrow$ FC($2048 \rightarrow m$) $\times 2$ ($\mu, \log \sigma^2$)
 - Decoder: FC($m \rightarrow 2048$) $\rightarrow$ ConvT($128 \rightarrow 64$) $\rightarrow$ ConvT($64 \rightarrow 32$) $\rightarrow$ ConvT($32 \rightarrow 3$) $\rightarrow$ Sigmoid
- $n_{\text{train}} = 20,000$, 150 epochs, KL cyclic annealing ($\beta_{\max} = 4$, 3 cycles)
- $m \in \{16, 32, 64, 128, 256\}$, seeds $\in \{42, 123, 777\}$

Table 20. Exp-Nat-Cifar and Exp-Nat-SVHN: Conv-VAE results for CIFAR-10 (left) and SVHN (right) (mean $\pm$ std, 3 seeds; KL cyclic annealing, 150 epochs, 3 cycles). At $m = 64$, CIFAR-10 latent TwoNN ID converges to the raw-data value (≈ 25), consistent with Pope et al. [16] ($d_{\text{ID}} \approx 26$).

m	CIFAR-10 (Exp-Nat-Cifar)		SVHN (Exp-Nat-SVHN)	
	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)	AU (mean $\pm$ std)	TwoNN ID (mean $\pm$ std)
16	16.0 $\pm$ 0.0	13.12 $\pm$ 0.12	16.0 $\pm$ 0.0	11.30 $\pm$ 0.12
32	32.0 $\pm$ 0.0	19.26 $\pm$ 0.23	32.0 $\pm$ 0.0	15.08 $\pm$ 0.08
64	64.0 $\pm$ 0.0	25.21 $\pm$ 0.41	64.0 $\pm$ 0.0	19.97 $\pm$ 0.12
128	128.0 $\pm$ 0.0	28.83 $\pm$ 0.41	125.3 $\pm$ 0.5	24.60 $\pm$ 0.36
256	256.0 $\pm$ 0.0	30.91 $\pm$ 0.64	206.7 $\pm$ 1.2	27.08 $\pm$ 0.34
Raw TwoNN=25.20, MLE=26.56 ($\approx$ Pope et al. estimate of 26)			Raw TwoNN=10.09, MLE=18.85	
$m^{\text{knee}} (\tau = 0.25)$: all seeds=256 (no knee in range)			m^{knee} : all seeds=256 (no knee in range)	

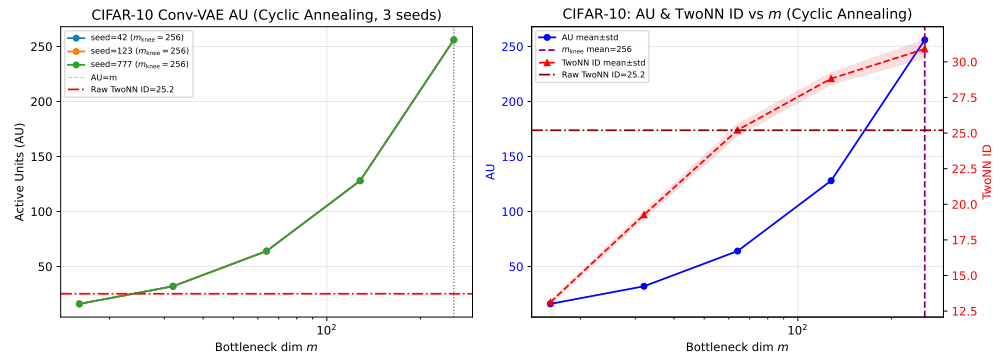


Figure 10. Exp-Nat-Cifar: m -dependence of AU and TwoNN ID for CIFAR-10 Conv-VAE (KL cyclic annealing, 3 seeds). TwoNN ID shows convergence consistent with Pope et al.[16] ($d_{\text{ID}} \approx 26$).

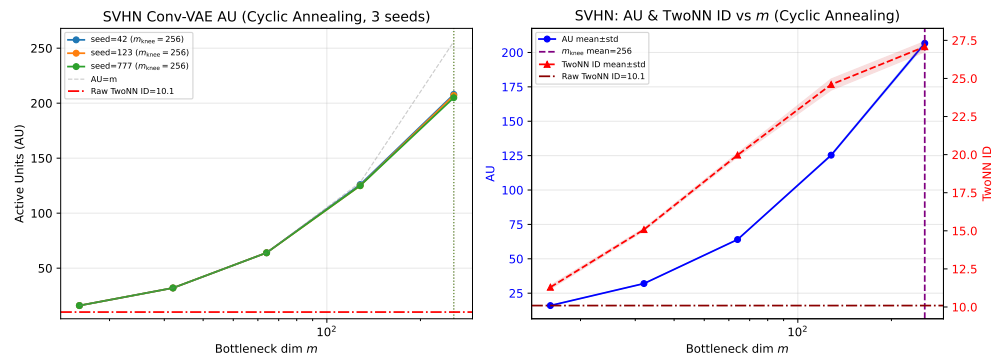


Figure 11. Exp-Nat-SVHN: m -dependence of AU and TwoNN ID for SVHN Conv-VAE (KL cyclic annealing, 3 seeds).

The primary aim of this experiment is limited to verifying whether the TwoNN ID attains the same scale as the values reported by Pope et al. within an appropriate bottleneck range (stable identification on the AU/m^{knee} side is not achieved for natural images; this experiment delimits the applicability boundary). The two main findings are as follows (Table 20). **CIFAR-10:** the latent TwoNN ID at $m = 64$, 25.21 ± 0.41 (3 seeds), is on the same scale as the raw-data TwoNN ID (25.20) and the value reported by Pope et al. [16] (≈ 26), but rises to 28.8–30.9 for $m \geq 128$ —the numerical consistency of TwoNN is restricted to a range of roughly $m \approx 2d_{\text{ID}}$. AU satisfies $AU = m$ for $m \leq 256$ (no saturation; m^{knee} at the grid boundary for all seeds), and we do not claim generalization of the entire framework to natural images. **SVHN:** $AU < m$ occurs for $m \geq 128$ but no AU saturation plateau is reached, and the latent TwoNN ID (27.08 at $m = 256$) diverges strongly from the raw-data TwoNN (10.09)—one instance of the applicability limits, showing that when AU uses all dimensions, geometric distortion of the latent space can lead to ID overestimation.

5.9. Exploring AU Saturation Conditions in CIFAR-10 Conv-VAE (Experiment Exp-NatImg-AU; Details in Appendix F)

This section and §5.11 are “quantification of the applicability boundary” experiments—rather than reporting a failure of stable AU/m^{knee} identification, they systematically identify the boundary conditions under which AU diagnostics become difficult. To disentangle the possible causes of the absence of AU saturation in Exp-Nat-Cifar ($\beta_{\text{max}} = 4$)—insufficient β , insufficient training, or insufficient capacity—we systematically varied $\beta_{\text{max}} \in \{10, 20, 40\}$ and two architectures (Standard/Deep) (CIFAR-10, $m \in [32, 384]$, KL cyclic annealing, 200 epochs, **single seed 42**; full setup, AU table, and figure in Appendix F, Table A7).

Key observations: (1) increasing β_{max} suppresses dimensions ($\beta_{\text{max}} = 40$, Deep: $AU = 208$ at $m = 384$, utilization 54%), but the $AU(m)$ curve continues to increase monotonically

under all conditions and no saturation plateau appears ($m^{\text{knee}} = 384$, the grid boundary, everywhere). (2) The latent TwoNN ID keeps increasing beyond the pixel-space TwoNN (25.30) (Arch-A, $\beta_{\text{max}} = 10$: 29.2 at $m = 256$), showing a tendency for the ID to be overestimated through latent-geometry distortion while AU uses all dimensions. (3) From the perspective of Theorem 1, the powerful convolutional decoder makes the effective noise variance τ_{eff}^2 far smaller than for MNIST, so increasing β alone hardly reverses the activation condition $\lambda_j > \beta \tau_{\text{eff}}^2$. We thus identified that identifying m^{knee} via AU saturation requires, as a necessary condition, either (a) exploration over a substantially wider m grid ($m \gg 2\hat{d}_{\text{ID}}$) or (b) a setting with restricted decoder capacity (§6.6). The extended sweep in the next subsection quantifies this boundary further.

5.10. Real-Data Geometry Comparison Experiment: Empirical Validation of the TwoNN Stability Condition (RQ2 and RQ3)

As an experimental validation of Proposition 4, this experiment (Exp-Real-Geom) sweeps **four types of models with systematically varied isometry** over multiple values of m on MNIST and Fashion-MNIST, and compares TwoNN, MLE, κ , Trustworthiness, and kNN overlap.

Experimental Setup (Exp-Real-Geom)

Models: we compare the following four types.

Standard AE no Jacobian regularization ($\kappa \approx 10^2$ – 10^7 , growing with m ; Table A5)

IsometricAE decoder-Jacobian regularization ($\kappa \approx 1.1$ – 1.4 ; Kato et al. [13])

CAE encoder-side contractive regularization (intermediate κ ; Rifai et al. [22])

DAE input-noise regularization (noise scale $\sigma = 0.1$; Alain et al. [23])

m -sweep: $m \in \{d, 2d, 3d, 5d, 10d\}$ (with $d = \hat{d}_{\text{ID}}$ set to 10 for MNIST and 9 for Fashion-MNIST; that is, $m \in \{10, 20, 30, 50, 100\}$ and $m \in \{9, 18, 27, 45, 90\}$, respectively).

Metrics: TwoNN ($k = 2$), MLE ($k = 10$), κ (decoder-Jacobian condition number), Trustworthiness ($k = 10$), and kNN overlap (kNN neighborhood-preservation rate, $k = 10$).

Data and training: MNIST and Fashion-MNIST, 20,000 training samples, seeds $\{42, 123, 777\}$ (3 seeds), 300 epochs, hidden layers $[512, 256]$, Adam ($\text{lr} = 10^{-3}$).

Experimental Results (Exp-Real-Geom)

Table 21 summarizes the main results (the full table over all m values and all metrics is given in Appendix E, Table A5).

Table 21. Exp-Real-Geom (summary): MNIST four-model comparison (mean $\pm$ std, 3 seeds, 300ep). TwoNN ID converges as $m/\hat{d}_{\text{ID}}$ increases in all four models, largely independently of κ . Full table in Appendix E.

Model	TwoNN ($m=d$)	TwoNN ($m=5d$)	TwoNN ($m=10d$)	κ ($m=10d$)
AE	7.3 ± 0.4	10.0 ± 0.1	10.0 ± 0.1	4.1×10^7
IsometricAE	7.5 ± 0.4	10.1 ± 0.1	10.1 ± 0.1	7.8
CAE	7.2 ± 0.5	9.9 ± 0.1	10.0 ± 0.1	3.2×10^3
DAE	7.6 ± 0.4	10.0 ± 0.1	10.0 ± 0.1	6.3×10^3

Discussion: $m \gg \hat{d}_{\text{ID}}$ Is the Dominant Condition

(1) Increasing $m/\hat{d}_{\text{ID}}$ is the dominant factor in TwoNN stabilization: across all four models, the TwoNN ID exhibits a common convergence pattern from $m = d$ (≈ 7.3 – 7.6) to $m = 5d$ – $10d$ (≈ 10.0 – 10.1). Inter-model differences at the same m remain within about ± 0.3 , far smaller than the improvement obtained by increasing m (about 2.5 points).

(2) The isolated effect of isometry improvement (κ reduction) is small: IsometricAE achieves $\kappa \approx 3\text{--}8$ (orders of magnitude lower than the $10^2\text{--}10^7$ of the AE), yet its TwoNN ID differs from that of the AE at the same m by no more than ± 0.2 . Improvements in Trustworthiness and kNN overlap are likewise driven by m (inter-model difference < 0.003 ; Appendix E), consistent with the implication of Proposition 4 that “ δ_T (the neighborhood rank-inversion rate) dominates TwoNN stability over ε (distortion).” A similar pattern was observed on Fashion-MNIST ($\hat{d}_{ID} = 9$; Appendix E).

Implications for real data: this experiment extends the findings of Exp-Iso-ID (synthetic manifolds; §5.2.4) to real data and multiple models, reinforcing the design guideline that “ m -grid design ($m \gg \hat{d}_{ID}$) is more effective than isometry bias mitigation for the practical stabilization of TwoNN estimation,” and supports the rationale for using a reference AE with $m_{\text{ref}} = 256 \gg \hat{d}_{ID} \approx 10$ in Stage 1.

5.11. Natural-Image Extended Sweep and the Stable Identification Criterion (RQ1)

In this section, we use the five stable-identification criteria defined in §3.4 to judge the degree to which AU/m^{knee} diagnostics are achieved on natural images.

Experimental Setup and Key Findings (Exp-NatImg-Sweep)

Substantially extending the previous section (Exp-NatImg-AU), we conducted an exhaustive sweep combining $m \in [32, 1024]$, $\beta_{\text{max}} \in \{10, 20, 40, 80\}$, decoder capacity (Standard/Deep/Thin), 300 epochs, and 3 seeds (details in Appendix F).

Key findings: the observation, under the conditions of this study, of plateau-like behavior with $AU \approx 24$ over $m \in [512, 1024]$ in the $\beta_{\text{max}} = 80$ /thin-decoder condition provides preliminary evidence that “AU saturation can occur even for natural images under sufficiently large β and restricted decoder capacity.” Across all conditions, the TwoNN ID remains stable at $24.7\text{--}25.4$ ($\sigma \leq 0.6$), consistent with the value reported by Pope et al. [16] (≈ 26). We found that stable identification of AU/m^{knee} requires larger m , higher β , and decoder-capacity control. The detailed evaluation of the five conditions (stable identification criteria; §3.4) is summarized in Appendix F. Furthermore, in the detection-method comparison (Exp-NatImg-Knee), the post-smoothing curvature method was the most reproducible, but it was also confirmed that stabilizing AU saturation itself takes precedence over improving the detection method (Appendix F).

6. Discussion

6.1. Answer to RQ1 and the Positioning of m^{knee}

Summary answer to RQ1: For RQ1—“under what grid-density, seed-count, and architecture conditions can m^{knee} be used as a diagnostic aid consistent with TwoNN/MLE ID estimates?”—the answers are as follows.

- **Identification of stabilization conditions (with explicit scope):** In the MNIST multi-seed sparse-grid experiments, m^{knee} fluctuated widely between 32 and 64, revealing that on coarse grids it cannot be used as a reliable diagnostic aid. Under a dense grid placed around the TwoNN estimate, with independent initialization and multiple seeds (Exp-Mnist-Dense-300), $m^{\text{knee}} = 10.3 \pm 0.9$ stabilized in the vicinity of the TwoNN ID (≈ 10). However, this result is confirmed only under the conditions **MNIST FC-VAE**, $\beta = 4$, **TwoNN-guided dense grid**, **three seeds**, and since the search region itself depends on the TwoNN estimate, it is not an independent validation (§5.3.6). The default description of m^{knee} is therefore “a TwoNN-guided exploratory change point,” and we do not recommend its general use without stating these conditions.

- **Applicability limits:** For the Conv-VAE ($\beta_{\max} = 4$) and natural images, stable identification was not reached within the tested range; conditions such as $\beta_{\max} \geq 10$, a dense grid, and control of decoder capacity are additionally required.

The instability of m^{knee} is not a side issue merely mitigated by grid density; we treat it as a **first-class limitation** of the framework (§6.7). The main contribution of this paper is the establishment of a diagnostic framework that places two stable indicators—the AU value (robust indicator) and the TwoNN ID (numerically stable auxiliary reference)—at its core, and positions m^{knee} as an exploratory complementary indicator (Table 2).

6.2. An Optimization-Dynamics Interpretation of m^{knee} Instability and Stabilization Strategies

Interpretation in terms of optimization dynamics: the gradient-competition hypothesis:

Proposition 2 formally established the grid sensitivity, but in actual training there exist additional sources of instability. The gradient of the β -ELBO is a linear combination of the reconstruction term $\nabla \mathcal{L}_{\text{rec}}$ and the KL term $\nabla \mathcal{L}_{\text{KL}}$, and these give rise to **gradient competition** near $m \approx d^*(\beta)$ (around the effective-dimensionality threshold):

- Reconstruction gradient ($\nabla \mathcal{L}_{\text{rec}}$): even for surplus dimensions $j > d^*$, there is a small reconstruction contribution in early training (a positive gain whenever $\lambda_j > 0$), which pushes toward activating the dimension.
- KL gradient ($\beta \nabla \mathcal{L}_{\text{KL}}$): a pressure in the opposite direction that pushes the same dimension back toward the prior.
- The speed at which this competition is resolved depends on the random seed (initialization), the m -grid (an independent optimization trajectory for each m), and the value of β . The convergence path toward the global optimum of Theorem 1 is not unique in non-linear optimization, and in low-sharpness loss landscapes the slope of the AU curve near $m \approx d^*$ is sensitive to initial conditions.

Cyclical KL annealing [2] increases β in stages, thereby gradually promoting the resolution of this gradient competition and improving convergence toward the global optimum (experiment Exp-Conv-Ann; §5.5.1).

Practical approaches to stabilizing m^{knee} :

Based on the findings of this study (including the grid-density $\times$ epoch factor separation in the Exp-Mnist-Dense experiment), we propose the following recommendations for using m^{knee} in actual design:

1. **Dense grid design (TwoNN-guided local refinement):** place dense step-1 points around the estimate $\hat{d}_{\text{ID}}$ (obtained from the TwoNN ID), as demonstrated in Exp-Mnist-Dense (§5.3.6). By Proposition 2, the grid spacing directly becomes the maximum error of m^{knee} . Note that the search region depends on the TwoNN estimate and that the seed factor remains.
2. **Ensemble m^{knee} :** compute each $m^{\text{knee}(s_i)}$ from the AU curves for multiple seeds and adopt the **median**:

$$m^{\text{knee,ens}} = \text{median}\{m^{\text{knee}(s_1)}, \dots, m^{\text{knee}(s_S)}\} \quad (9)$$

The median is robust to outliers in which m^{knee} is stuck at the grid boundary (when S is even, we take the arithmetic mean of the two adjacent values).

3. **Adopting KL annealing and extending the m -grid:** For the Conv-VAE, annealing suppresses unnecessary collapse at large m , but observing AU saturation requires extending the m -grid to at least twice the estimated $d^*(\beta)$ (e.g., $m > 64$ for the MNIST Conv-VAE; §5.5.1).

4. **Cross-checking against stable indicators:** estimates of m^{knee} should always be cross-checked against the AU value (robust indicator) and the TwoNN ID (numerically stable auxiliary reference), with the final judgment based on the consistency of the three indicators (see Table 22).
5. **Second difference of the AU curve (alternative method):** when a fixed threshold τ fails (e.g., SVHN), use the maximum-curvature method that takes as m^{knee} the point of maximum absolute discrete second difference $\Delta^2 \text{AU}(m_i)$ of $\text{AU}(m)$ (also validated in parallel in the Exp-Mnist-Dense experiment; see Appendix C).

The rationale for the choice $\tau_{\text{knee}} = 0.25$, its sensitivity analysis (Exp-Tau-Cross), and the division of labor between the τ method and the second-difference method are collected in Appendix C. In brief, under dense-grid conditions (Exp-Mnist-Dense) the τ method (10.3 ± 0.9) is more robust than the second-difference method (18.3 ± 6.8); when the τ method becomes unstable (the AU growth rate staying near the boundary value across multiple m), switching to the second-difference alternative is effective.

6.3. Numerical Stability and Systematic Bias of the TwoNN ID: Distinguishing “Stability” from “Accuracy”

Experiments Exp-Stab and Exp-Geom showed that the TwoNN ID is “numerically stable (low variance),” but this is a claim distinct from being “unbiased (accurate) with respect to the true intrinsic dimension.” Making this distinction explicit is important for operating our framework. Through Proposition 4 (§4.2), we presented the semi-theoretical framework that “preservation of neighborhood ranks (δ_T) rather than isometry (ε) is the dominant factor for TwoNN stability,” and demonstrated it on real data across multiple models in experiment Exp-Real-Geom (§5.10).

What numerical stability means: In experiment Exp-Geom (MNIST AE m -sweep), for $m \geq 48 \approx 5\hat{d}_{\text{ID}}$ the TwoNN ID converges stably within the range 10 ± 0.5 , and even against the sharp increase of κ at $m = 96\text{--}128$ ($9.94 \rightarrow 37.0$) the variation remains small (§5.2.4, Table A2). That is, TwoNN is stable in the sense that “the reference AE outputs nearly the same value even when m is changed.”

Possibility of systematic bias: However, when the reference AE embeds the space with distortion ($\kappa \gg 1$), TwoNN estimates “the local degrees of freedom of the reference AE’s latent manifold,” which may deviate systematically from the intrinsic dimension of the true data manifold. This is not inconsistent with stably returning a “particular value”—it is, so to speak, a “stably biased estimate.” We therefore treat TwoNN not as an unbiased estimator of the true d_{ID} but as an **auxiliary reference estimator**.

Grounds for $\hat{d}_{\text{ID}} \approx 10\text{--}12$ on MNIST: As independent evidence reinforcing the meaning of the TwoNN estimate, the following multiple observations are mutually consistent:

- MSE elbow (AE): the rapid MSE decrease slows down around $m \approx 16\text{--}20$ (a value somewhat larger than d_{ID} , but of the same order; experiment Exp-Mnist-Base).
- PCA spectrum: 43 dimensions at 80% cumulative variance and 86 dimensions at 90% (indicating $d_{\text{ID}} \ll 43$ as an upper bound on the non-linear dimension; experiment Exp-Mnist-PCA).
- Consistency check on CIFAR-10: on CIFAR-10, the latent TwoNN ID of 25.21 ± 0.41 was confirmed to be on the same scale as the value reported by Pope et al. [16] (≈ 26), supporting the numerical stability of TwoNN on natural images as well (experiment Exp-Nat-Cifar).
- Agreement with MLE: for $k = 5, 10, 20$, $\text{MLE} = 11.5\text{--}12.4$ and $\text{TwoNN} = 10.0\text{--}10.5$, i.e., estimators based on different principles yield values of the same order.

These multiple pieces of evidence are mutually consistent and raise the plausibility that $\hat{d}_{\text{ID}} \approx 10\text{--}12$ for MNIST is not an artifact specific to the reference AE, but they do not

directly prove that TwoNN is geometrically unbiased with respect to the true d_{ID} . That is, the TwoNN ID in this paper is a **stable latent-space reference** (target quantity (ii); §3.1), not an estimate of the true intrinsic dimension of the input distribution (target quantity (i)) itself. A stably biased estimator can still guide a diagnostic pipeline, but quantifying the deviation remains open: we leave as future work (a) a systematic comparison with ID estimators based on different principles, such as DANCo [15] and ESS, and (b) reporting bootstrap confidence intervals in addition to across-seed standard deviations.

6.4. The Mechanism by Which AU Dynamics Reflect the Effective Representation Dimensionality

Choosing between AU_{sat} and m^{knee} . For simply connected synthetic manifolds (Swiss Roll, etc.), a sharp AU saturation ($\text{AU}_{\text{sat}} = d_{\text{true}}$) is observed, so AU_{sat} serves as a direct readout of the effective representation dimensionality. By contrast, for complex distributions such as real data (multiple classes, disconnected submanifolds), AU increases monotonically and gradually without exhibiting a clear saturation. In this case, m^{knee} —as “the point where KL regularization begins to act effectively”—provides a **rough practical guide** to the effective representation dimensionality (Table 22; note that the precise reproducibility of the numerical value depends on the m -grid and the seed).

Table 22. Guidelines for using AU_{sat} vs. m^{knee} (based on this study).

Data type	AU pattern	Recommended indicator
Simple synthetic manifolds (simply connected)	Sharp saturation	AU_{sat} (consistent with d_{true})
Topologically non-trivial synthetic manifolds	Stepwise saturation	AU_{sat} (tends to match d_{emb} , conditional)
Real data (MNIST, Fashion-MNIST)	Gradually increasing	m^{knee} (rough heuristic; grid/seed-dependent)

6.5. Incidental Findings: Topological Structure, Training Dynamics, and Geometric Regularization
Topological structure and AU_{sat} (experiment Exp-Syn-Top)

For the Torus and the Möbius band, $\text{AU}_{\text{sat}} = d_{\text{emb}} = 3 (> d_{\text{ID}} = 2)$ was observed, suggesting a correspondence with the topological constraint arising from the contractibility of the VAE’s Gaussian prior [10,11]. However, this correspondence is conditional on standardization and sufficient convergence, and it does not apply to $\text{AU}_{\text{sat}} = 36 \gg \hat{d}_{\text{ID}}$ on MNIST (experiment Exp-Mnist-Ent is preliminary; see §6.7 for details). This finding is an incidental observation independent of the framework’s main conclusion (reading the effective dimensionality via m^{knee}), and we leave it as future work.

Training dynamics (experiment Exp-Mnist-Dyn, single seed)

The two-phase training dynamics on MNIST ($m = 16, 200$ epochs)—rapid formation of global structure followed by a gradual increase in the local effective dimensionality—is an exploratory single-seed observation, and multi-seed validation is left as future work.

Geometric regularization and TwoNN stability (experiments Exp-Syn-Iso and Exp-Iso-ID; auxiliary experiments for RQ3)

The focus of RQ3 is “the effect of geometric regularization and bottleneck margin on TwoNN stability.” Experiment Exp-Iso-ID (§5.2.4) revealed that improving isometry within $\kappa \in [1.1, 2.1]$ does not significantly reduce the systematic bias of the TwoNN ID—the main cause of TwoNN underestimation is an insufficient bottleneck margin ($m < d_{\text{true}}$), and even though the IsometricAE ($\kappa \approx 1.04$) achieves isometry (experiment Exp-Syn-Iso), its contribution to TwoNN accuracy is small (§6.7). The primary usefulness of the IsometricAE is as an “isometry-verification tool,” not as a primary readout indicator of the effective dimensionality. The cost of computing the decoder Jacobian (a factor of $O(m)$) also limits its applicability to large-scale data (§6.7).

6.6. Theoretical Considerations on Delayed AU Saturation in Conv-VAEs

For the FC-VAE, the AU growth slows down near $m \geq \hat{d}_{\text{ID}} \approx 10$, and m^{knee} was qualitatively consistent with $\hat{d}_{\text{ID}}$ (experiments Exp-Mnist-Base, Exp-Mnist-Dense). By contrast, for the Conv-VAE, $\text{AU} \approx m$ persisted over the entire range $m \leq 64$, and even with KL annealing (experiment Exp-Conv-Ann) no AU saturation plateau was observed within the tested range (experiment Exp-Conv-Fix). This “delayed AU saturation” can be understood via the following mechanism.

Interpretation via a reduced effective τ_{eff}^2 By Theorem 1 (the linear-Gaussian rate-distortion surrogate), the number of active dimensions is determined by $d^*(\beta) = |\{j : \lambda_j > \beta\tau_{\text{eff}}^2\}|$. Because a convolutional decoder has stronger spatial reconstruction capability than a fully connected decoder, the effective reconstruction noise variance τ_{eff}^2 is expected to be smaller under the same β . The smaller τ_{eff}^2 is, the lower the water-filling threshold $\beta\tau_{\text{eff}}^2$, so more eigenvalues satisfy the activation condition $\lambda_j > \beta\tau_{\text{eff}}^2$, and $d^*(\beta)$ takes a larger value (the water level is lower and more dimensions are judged effective). Whereas experiment Exp-Mnist-PCA estimates $\tau_{\text{eff}}^2 \approx 0.30$ for the FC-VAE, for the Conv-VAE we have $\tau_{\text{eff}}^2 \ll 0.30$, and $d^*(\beta)$ is expected to greatly exceed the tested range $m \leq 64$. Applying the framework to Conv-VAEs therefore requires substantially extending the m -grid until it reaches the estimated $d^*(\beta)$.

Experimental verification This interpretation is consistent with the results of §5.6 and §5.7: at $\beta_{\text{max}} = 4$, no saturation plateau appears even when extending to $m = 512$ ($d^*(\beta) > 512$), whereas increasing β_{max} to 10 and 20 yields identifiable $m^{\text{knee}} \approx 128$ and 96 (Table 19)—exactly the theoretical prediction that increasing β shifts $d^*(\beta)$ toward smaller values. The non-observation of saturation at $\beta_{\text{max}} = 4$ is thus not an applicability limit of the framework but a consequence of β being insufficient relative to the decoder capacity; application to Conv-VAEs requires strengthening β or reducing decoder capacity.

Implications for natural images For CIFAR-10 and SVHN, we confirmed that the TwoNN ID remains numerically stable on natural images as well, but stable identification on the $\text{AU}/m^{\text{knee}}$ side was not achieved (§5.8, §5.9). In experiment Exp-NatImg-AU ($m \in [32, 384]$, $\beta_{\text{max}} \in \{10, 20, 40\}$), $m^{\text{knee}} = 384$ (the grid edge) persisted under all conditions, and it was quantitatively identified that AU saturation requires $m \gg 2\hat{d}_{\text{ID}}$ together with $\beta_{\text{max}} \gg 40$ or a thin decoder. The observation of plateau-like behavior in Exp-NatImg-Sweep (§5.11) was achieved under these conditions ($\beta_{\text{max}} = 80$, thin decoder).

6.7. Limitations and Future Work

(A) Scope of the Evaluated Data and Generalizability

- **Conv-AE and Conv-VAE validation** (experiments Exp-Conv-AE, Exp-Conv-Fix, Exp-Conv-Ann, Exp-Conv-M256): We confirmed the TwoNN ID consistency of the Conv-AE (Exp-Conv-AE). At $\beta_{\text{max}} = 4$, no AU saturation plateau was observed up to $m \leq 512$, but the high- β experiment (Exp-ConvV-HiBeta) demonstrated that an insufficient β setting was the main cause (§5.7). The CIFAR-10 and SVHN experiments confirmed that the TwoNN ID is consistent with the values reported by Pope et al. [16] (§5.8), and the extended sweep (Exp-NatImg-Sweep, §5.11) conditionally observed AU plateau-like behavior; however, none of the five conditions for stable identification were met, and full application to large-scale natural images remains future work (in particular, achieving the β -stability and seed-stability conditions is the next goal).
- **Higher-order topological manifolds:** Extending the Exp-Syn-Top experiment to synthetic manifolds with higher-order topological invariants, such as the Klein bottle, remains an open task.
- **The instability of m^{knee} is a first-class limitation:** as the multi-seed validation (§5.3.4) shows, the sparse-grid instability of m^{knee} is not resolved by extending training, and its

dense-grid stabilization has been confirmed only under the restricted setting of MNIST FC-VAE, $\beta = 4$, and a TwoNN-guided search region. Grid density and multiple seeds cannot in general be regarded as sufficient stabilization measures, and we do not recommend using m^{knee} without stating these conditions.

- **Remaining single-seed experiments:** Exp-Mnist-Base, Exp-Mnist-Dyn, Exp-Mnist-Ent, Exp-Fashion, Exp-ConvV-HiBeta, Exp-NatImg-AU, and others are single-seed (marked explicitly in Appendix H, Table A10); statements based on them are restricted to preliminary observations. Multi-seed replication remains future work.

(B) Sensitivity of Indicators and Hyperparameters

- **Stage 1 ID estimation: rigorous assessment of isometry bias:** In this paper, the TwoNN ID is estimated via the latent representation of a reference AE ($m_{\text{ref}} = 256$) (Stage 1). **This estimate is inherently an approximation and contains the following bias sources:**

- (i) **Lack of isometry:** the condition number $\kappa \approx 10^9$ of the $m = 256$ reference AE deviates from isometry ($\kappa = 1$) by orders of magnitude (Proposition 3; experiment Exp-Syn-Iso). A distance-based TwoNN ID in a latent space lacking isometry can lead to under- or over-estimation of the true intrinsic dimension.
- (ii) **Synergy with KL non-saturation:** in experiment Exp-Conv-Fix, the TwoNN ID of the Conv-VAE under AU non-saturation ($\text{AU} = m$) was 13.10, substantially inflated relative to the FC-VAE under AU saturation (10.03), showing that isometry bias and AU non-saturation synergistically distort the ID estimate.
- (iii) **Possible circularity:** estimating $\hat{d}_{\text{ID}}$ via the reference AE and then using that value as the evaluation criterion for m^{knee} can create circularity between estimation and evaluation.

Mitigations: We propose using the IsometricAE ($\kappa \approx 1$) as the reference AE, or verifying that the TwoNN ID converges across multiple m_{ref} (e.g., $m \in \{64, 128, 256\}$). In this paper we confirmed that the TwoNN ID stabilizes asymptotically from $m = 64$ onward (10.03 ± 0.12), which is the practical basis for the $\hat{d}_{\text{ID}}$ estimate (§5.2.4). A rigorous quantification of the isometry bias (including comparison with DANCo-type estimators and bootstrap confidence intervals; §6.3) remains future work.

- **Non-universality of τ_{knee} :** For Fashion-MNIST, a boundary sensitivity in which m^{knee} changes around $\tau = 0.25$ has been confirmed, and we recommend combined use with the second-difference alternative method (Appendix C).
- **Preprocessing dependence of AU_{sat} :** AU_{sat} varies with the normalization method (experiment Exp-Syn-Top). Reproducing $\text{AU}_{\text{sat}} \approx d_{\text{emb}}$ requires standardization (zero mean, unit variance).
- **Dependence on β :** From preliminary experiments (50 epochs, $\beta \in \{1, 2, 4, 8\}$), $\text{AU} \approx m$ (under-regularization) was observed at $\beta = 1$, and signs of over-regularization at $\beta = 8$ (Appendix D, Figure A1). The stability of the AU pattern for $\beta \in [2, 8]$ has been confirmed, but the precise value of m^{knee} also depends on the m -grid and the random seed (§6.1).

(C) Computational Cost

- Due to the Jacobian computation cost of the IsometricAE (a factor of $O(m)$), direct application to CIFAR-10 ($\approx 64\times$) and ImageNet ($\approx 3,000\times$) is difficult.
- A full m -sweep (11 dimensionalities $\times$ 300 epochs $\times$ 2 models) takes several hours to several days.

7. Conclusions

In this paper, we proposed a **multi-indicator framework** for diagnosing the effective latent dimensionality of AEs/VAEs. The central contribution of this paper is a clear role classification of AU values (robust indicator), TwoNN ID (numerically stable auxiliary reference estimator), and m^{knee} (exploratory indicator), quantifying the reliability and applicability conditions of each indicator from both the experimental and theoretical sides (Table 2). On the theoretical side, we derived the AU activation threshold $\lambda_j > \beta\tau^2$ under the linear-Gaussian rate–distortion surrogate from the water-filling theorem and the KKT conditions (Theorem 1), and presented a grid-sensitivity upper bound for m^{knee} (Proposition 2) together with an ensemble stabilization strategy (Section 6.2). On the experimental side, we confirmed the high reproducibility of AU values ($\sigma \leq 0.5$) and TwoNN ID (10.03 ± 0.12) on MNIST (3 seeds), and stabilized $m^{\text{knee}} = 10.3 \pm 0.9$ under a dense grid placed around the TwoNN estimate, with independent initialization and multiple seeds (Exp-Mnist-Dense-300; Section 5.3.6). This stabilization is a **TwoNN-guided local refinement**, and its verified scope is limited to MNIST FC-VAEs with $\beta = 4$, a dense grid, and three seeds—we do not recommend general use of m^{knee} without stating these conditions. We confirmed the same tendency on Fashion-MNIST with a single seed (Section 5.4). Through the real-data geometry comparison experiment (Exp-Real-Geom, Section 5.10), we supported the claim that “ $m \gg \hat{d}_{\text{ID}}$, rather than isometry improvement, is the dominant condition for TwoNN stability” by a four-model comparison on MNIST (Table 21) and a qualitative confirmation on Fashion-MNIST (Appendix E) (with theoretical grounding in Proposition 4). The TwoNN ID is a stable latent-space reference, and its unbiasedness with respect to the true intrinsic dimension of the input distribution is not guaranteed (Section 6.3).

We organize the applicability of the framework into three levels (Table 23; evidence: Sections 5.3.4, 5.3.6, 5.5.1, and 5.11).

Table 23. Three-tier applicability of the diagnostic framework (summary of evidence).

Setting	AU values	TwoNN ID	m^{knee} / requirements
FC-VAE (MNIST, 3 seeds)	Stable ($\sigma \leq 0.5$)	Stable ($\sigma \leq 0.12$, $m = 64$)	Exploratory guide near TwoNN ID only under dense grid + independent initialization + multiple seeds (Exp-Mnist-Dense-300: 10.3 ± 0.9); Fashion-MNIST: same tendency, single seed
Conv-VAE (MNIST)	Stable; $\text{AU} < m$ from $m \geq 96$	Stable (inflates while $\text{AU} = m$)	Identifiable with $\beta_{\text{max}} \geq 10$ ($m^{\text{knee}} \approx 96\text{--}128$); at $\beta_{\text{max}} = 4$, $d^*(\beta) > 512$ (Exp-Conv-M256/M512, Exp-ConvV-HiBeta)
Natural images (CIFAR-10/SVHN)	Plateau-like only at $\beta_{\text{max}} = 80$ + thin decoder	Consistent with Pope et al. [16] at $m \approx 2\hat{d}_{\text{ID}}$	Stable identification: all five criteria unmet; future work (Exp-Nat-Cifar, Exp-NatImg-Sweep)

We position the Conv-VAE and natural-image experiments (Sections 5.5.1–5.11) not as a “success or failure” verdict but as a validation that quantifies the boundary conditions under which AU diagnostics become difficult. In the high- β experiment (Exp-ConvV-

HiBeta), AU saturation was observed for $\beta_{\max} \geq 10$, consistent with the interpretation that the non-saturation at $\beta_{\max} = 4$ is attributable to the β setting, and in the CIFAR-10 extended sweep (Exp-NatImg-Sweep) we observed AU plateau-like behavior under our experimental conditions with $\beta_{\max} = 80$ and a thin decoder (all five stable-identification conditions remained unmet; Appendix F). The main directions for future work are: (i) achieving stable identification on natural images (a dense sweep over $\beta_{\max} \in [40, 120]$ and improved seed stability); (ii) extending Theorem 1 to nonlinear settings (analysis based on local linearization); and (iii) scaling to large datasets (ImageNet, with reduced computational cost).

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Data Availability Statement: MNIST [26], Fashion-MNIST [27], CIFAR-10 [3], and SVHN [4] are publicly available. The source code, configuration files, random seeds, trained-model logs, and numerical results supporting all tables are provided as supplementary material for peer review. Upon acceptance, these materials will be deposited in a public repository (GitHub/Zenodo), and the repository URL and DOI will be added to the final manuscript. The repository will include the scripts required to reproduce the AU, TwoNN/MLE, reconstruction-error, and grid-sensitivity analyses reported in this paper.

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Abbreviations

The following abbreviations are used in this manuscript:

AE	Autoencoder
VAE	Variational Autoencoder
AU	Active Units
ID	Intrinsic Dimension
TwoNN	Two-Nearest-Neighbor (ID estimator)
MLE	Maximum Likelihood Estimation
MSE	Mean Squared Error
PCA	Principal Component Analysis
KL	Kullback–Leibler (divergence)
FC	Fully Connected
CNN / Conv	Convolutional Neural Network
CAE	Contractive Autoencoder
DAE	Denoising Autoencoder
NSR	Noise Selectivity Ratio

Appendix A. Proof Sketches of Propositions

Proposition A1 (Effective rank of the decoder Jacobian). *When an AE with sufficient capacity reconstructs a manifold $\mathcal{M}$ exactly, the number of significant singular values of the decoder Jacobian $J_g(z)$ equals $\min(m, d_{\text{emb}})$.*

Proof sketch: Under exact reconstruction, the range of g spans the tangent space $T_{g(z)}\mathcal{M}$ of $\mathcal{M}$. If $\mathcal{M}$ is d -dimensional, then $\text{rank}(J_g) = d$, and the singular values of the surplus dimensions approach zero. (Finite capacity and optimization error are neglected.)

Proposition A2 (Noise-direction selectivity). *The optimal reconstruction function of a DAE removes noise in the normal directions of the manifold more selectively than noise in the tangent directions.*

Proof sketch: By Alain et al. [23], the DAE reconstruction direction satisfies $r(x) - x \approx \sigma^2 \nabla \log p(x)$, and $\nabla \log p(x)$ points strongly in the normal directions of the manifold (under the manifold hypothesis). (The asymptotically optimal solution is assumed; approximation errors under finite noise and finite capacity are neglected.)

Proposition A3 (Neighborhood-structure preservation and the optimal bottleneck). *At $m = d_{\text{true}}$ (or d_{emb}), Trustworthiness improves sharply toward its maximum, and further increases in m yield only limited improvement.*

Proof sketch: By Proposition 1, a continuous injection is impossible for $m < d$, so collisions of information are unavoidable and Trustworthiness degrades. For $m \geq d$ the constraint is lifted, and for $m > d$ the entire manifold structure is already representable, so the effect of additional dimensions is limited. (Finite samples and finite capacity are neglected.)

Appendix B. Detailed Derivation of Theorem 1

In this appendix, we detail the algebraic steps omitted from the proof sketch of Theorem 1.

Remark A1 (Assumptions underlying the approximation). *The expression $L_j = -D_j/(2\tau^2) - \beta R_j$ for the per-dimension β -ELBO contribution used in this derivation implicitly involves a rate-distortion surrogate approximation that identifies the expected KL divergence of the VAE, $\mathbb{E}_x \text{KL}(q(z_j|x)||p(z_j))$, with the mutual information $R_j = I(z_j; y_j)$. Since the average KL of an actual VAE also includes the mismatch between the aggregated posterior and the prior, this derivation should be interpreted as an “AU activation condition under the linear-Gaussian rate-distortion surrogate” (as stated explicitly in the assumptions of Theorem 1).*

Step 1: Derivation of the Optimal Decoder Weight

As a scalar problem, for direction j consider $y_j \sim \mathcal{N}(0, \lambda_j)$ (the j -th principal component of the data), the encoder $q_j(z_j|y_j) = \mathcal{N}(a_j y_j, \sigma_j^2)$, and the decoder $\hat{y}_j = b_j z_j$. For the decoder weight b_j , we minimize the marginalized reconstruction MSE (with σ_j^2, a_j held fixed):

$$D_j(b_j) = \mathbb{E}_{y_j, z_j}[(y_j - b_j z_j)^2] = \lambda_j - 2a_j \lambda_j b_j + b_j^2(a_j^2 \lambda_j + \sigma_j^2). \quad (\text{A1})$$

Solving $\partial D_j / \partial b_j = 0$, the optimal decoder weight is

$$b_j^* = \frac{a_j \lambda_j}{a_j^2 \lambda_j + \sigma_j^2} =: \frac{a_j \lambda_j}{\rho_j}, \quad \rho_j := a_j^2 \lambda_j + \sigma_j^2. \quad (\text{A2})$$

Step 2: Expressions for the Optimal MSE and the Mutual Information

Substituting Equation (A2) into Equation (A1) yields:

$$D_j^* = \lambda_j - \frac{a_j^2 \lambda_j^2}{\rho_j} = \frac{\lambda_j \sigma_j^2}{\rho_j}. \quad (\text{A3})$$

Moreover, the channel capacity (mutual information) of $y_j \rightarrow z_j$ is:

$$I(z_j; y_j) = \frac{1}{2} \log \frac{\rho_j}{\sigma_j^2} = \frac{1}{2} \log \left(1 + \frac{a_j^2 \lambda_j}{\sigma_j^2} \right) =: R_j. \quad (\text{A4})$$

Step 3: Rate-Distortion Form of the β -ELBO

With $b_j = b_j^*$ held fixed, the β -ELBO contribution with respect to (σ_j^2, a_j) can be written as $L_j = -D_j/(2\tau^2) - \beta R_j$ (equivalence with KL minimization [1]). Taking ρ_j as the parameter and substituting $D_j^* = \lambda_j \sigma_j^2 / \rho_j$ and $R_j = \frac{1}{2} \log(\rho_j / \sigma_j^2)$, we obtain the expression for the β -ELBO contribution used in this appendix:

$$L_j = -\frac{\lambda_j \sigma_j^2}{2\tau^2 \rho_j} - \frac{\beta}{2} [\rho_j - 1 - \log \rho_j]. \quad (\text{A5})$$

(Here $\sigma_j^2 = \rho_j - a_j^2 \lambda_j$ and $\rho_j \geq \sigma_j^2$.)

Step 4: Derivation of the Optimal Distortion D_j^ via KKT Conditions*

Taking the distortion $D_j \in (0, \lambda_j)$ as the primal variable, we maximize $L_j = -D_j/(2\tau^2) - \beta R_j(D_j)$. From the Gaussian-channel rate-distortion relation $R_j(D_j) = \frac{1}{2} \log(\lambda_j / D_j)$:

$$\frac{\partial L_j}{\partial D_j} = -\frac{1}{2\tau^2} + \frac{\beta}{2D_j} = 0 \implies D_j^* = \beta\tau^2. \quad (\text{A6})$$

This corresponds to the “water level” $\mu = \beta\tau^2$ of Shannon’s water-filling theorem [1]. However, the solution in Equation (A6) is feasible (active) only when the feasibility constraint $D_j^* = \beta\tau^2 \leq \lambda_j$ is satisfied; when $\lambda_j < \beta\tau^2$, not encoding dimension j ($a_j = 0$, dead) is optimal.

Step 5: Computing the ELBO Values of the Active and Dead States

Dead: With $a_j = 0$, $\sigma_j^* = 1$ (KL = 0), and $D_j = \lambda_j$,

$$L_j^{(\text{dead})} = -\frac{\lambda_j}{2\tau^2}. \quad (\text{A7})$$

Active: Substituting $D_j^* = \beta\tau^2$, we have $R_j^* = \frac{1}{2} \log(\lambda_j / (\beta\tau^2))$, and therefore:

$$L_j^{(\text{active})} = -\frac{\beta\tau^2}{2\tau^2} - \beta \cdot \frac{1}{2} \log \frac{\lambda_j}{\beta\tau^2} = -\frac{\beta}{2} - \frac{\beta}{2} \log \frac{\lambda_j}{\beta\tau^2}. \quad (\text{A8})$$

Step 6: Exact Expression for the ELBO Gain ΔL_j and Sign Analysis

We compute the activation gain $\Delta L_j := L_j^{(\text{active})} - L_j^{(\text{dead})}$. Setting $t := \lambda_j / (\beta\tau^2) > 0$:

$$\Delta L_j = \left(-\frac{\beta}{2} - \frac{\beta}{2} \log t \right) - \left(-\frac{\beta\tau^2 \cdot t}{2\tau^2} \right) = \frac{\beta}{2} (t - 1 - \log t). \quad (\text{A9})$$

Analysis of the function $h(t) := t - 1 - \log t$:

- Since $h'(t) = 1 - 1/t = (t - 1)/t$, we have $h' = 0$ at $t = 1$ (the unique stationary point).
- Since $h''(t) = 1/t^2 > 0$, $t = 1$ is the **global minimum**, with $h(1) = 0$.
- For $t > 0$, $t \neq 1$, we have $h(t) > 0$ (strictly above the global minimum value 0).

Therefore, the sign of $\Delta L_j = (\beta/2) h(t)$ is:

$$\Delta L_j > 0 \iff h(t) > 0 \iff t \neq 1. \quad (\text{A10})$$

Step 7: Combining Feasibility and the Saturation Condition

The feasibility condition of the active solution ($D_j^* = \beta\tau^2 \leq \lambda_j$) requires $t \geq 1$. For $t < 1$ ($\lambda_j < \beta\tau^2$), the active solution is infeasible and the dead state is forced. At $t = 1$ ($\lambda_j = \beta\tau^2$), $\Delta L_j = 0$ (indifference). For $t > 1$ ($\lambda_j > \beta\tau^2$), the active solution is feasible and $\Delta L_j > 0$ (strictly preferred).

Summarizing the above:

$$\text{dimension } j \text{ is active} \iff \lambda_j > \beta\tau^2, \quad (\text{A11})$$

that is, the saturation-threshold condition (5) of Theorem 1 holds. $\square$

Remark A2. Near $t = 1$, the function $h(t) = t - 1 - \log t$ of Equation (A9) admits the second-order expansion $h(t) \approx (t - 1)^2/2$ (that is, $\Delta L_j \approx (\lambda_j - \beta\tau^2)^2/(2\beta\tau^4)$). The exact expression is Equation (A9); this approximation is a supplementary remark showing intuitively that the activation gain becomes quadratically small near the threshold $\lambda_j = \beta\tau^2$.

Appendix C. Sensitivity Analysis and Cross-Dataset Validation of τ_{knee} (Experiment Exp-Tau-Cross)

Rationale for Choosing $\tau_{\text{knee}} = 0.25$

Theoretical rationale: By Theorem 1, the raw growth rate $s(m) = \Delta \text{AU} / \Delta m$ of $\text{AU}(m)$ (Equation (2)) exhibits a theoretical step: $s \approx 1$ for $m < d^*(\beta)$ (before saturation; each new dimension becomes fully active) and $s \rightarrow 0$ for $m > d^*(\beta)$ (after saturation; additional dimensions remain inactive). Because the actual AU takes discrete integer values, this transition occurs smoothly (or discretely). The value $\tau_{\text{knee}} = 0.25$ corresponds to “at most 1/4 of the maximum growth rate” and, as a midpoint between the pre-saturation regime ($s \approx 1$) and the post-saturation regime ($s \approx 0$), constitutes a well-balanced threshold that captures the onset of the “clearly slowed” region.

Approximating the normalized growth rate $\rho(m)$ (Equation (3)) by an exponential saturation model gives

$$\rho(m) \approx \exp(-\gamma(m - d_{\text{ID}})). \quad (\text{A12})$$

We define the slowdown point as the point where $\rho(m) < \tau_{\text{knee}} = 0.25$ (below 25% after normalization) (Equation (4)).

As a more rigorous threshold choice, using the point where the discrete second difference of $\text{AU}(m)$ is maximized (the maximum-curvature point) is theoretically motivated, but due to the discreteness of AU it is often numerically unstable on real data. The value $\tau = 0.25$ serves as a stable surrogate near this maximum-curvature point (see the Exp-Tau-Cross sensitivity analysis).

Empirical rationale: On MNIST (the single-seed Exp-Mnist-Base experiment), $m^{\text{knee}} = 12$ (stable) over the entire range $\tau \in [0.1, 0.5]$. On Fashion-MNIST, $\tau \approx 0.25$ exhibits boundary sensitivity (Table A1). The existence of boundary sensitivity suggests that the gradient transition of the $\text{AU}(m)$ curve occurs at values near $\tau = 0.25$, which in turn corroborates the “discriminative power” of this threshold.

Cross-Dataset Validation (Experiment Exp-Tau-Cross)

Table A1 shows the cross-validation results on Fashion-MNIST and on Gaussian mixture distributions (5 and 10 components).

Table A1. Exp-Tau-Cross: Cross-dataset robustness verification of τ_{knee} (single run per dataset; protocol independent of Exp-Fashion, see text).

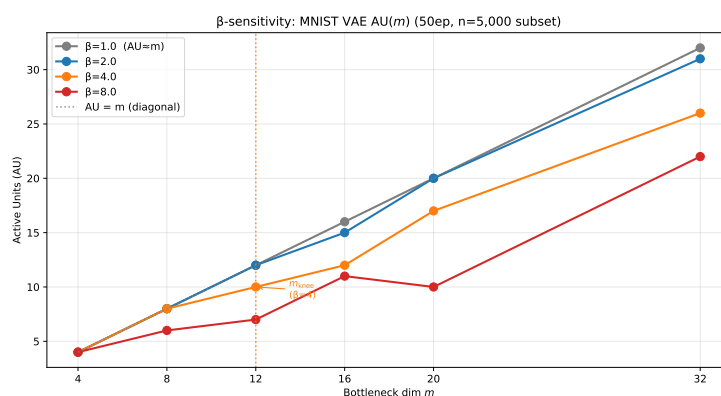
Dataset	AU _{sat}	TwoNN ID	$\tau = 0.1$	$\tau = 0.2$	$\tau = 0.25$	$\tau = 0.3$	$\tau = 0.4$	$\tau = 0.5$
Fashion-MNIST	10	7.00	20	20	20	8	8	8
GMix 5-comp.	1	4.27	2	2	2	2	2	2
GMix 10-comp.	3	3.02	2	2	2	2	2	2

On Fashion-MNIST, boundary sensitivity was confirmed at the boundary between $\tau = 0.25$ and 0.30 , where m^{knee} changes from 20 to 8 (because multiple values of m exist at which the AU growth rate coincides with the boundary value 0.25). In this case, we recommend using the second-difference alternative (see the stabilization strategies in §6.2).

Note that the Fashion-MNIST values in Table A1 (AU_{sat} = 10, TwoNN ID = 7.00, and $m^{\text{knee}} = 20$ at $\tau = 0.25$) were obtained under a setup dedicated to this cross-validation (150 epochs, a sparse cross-validation grid, and an independent training run), which differs from the Exp-Fashion setup of §5.4 (100 epochs, extended grid up to $m \leq 256$). The absolute values are therefore not directly comparable across the two, and the purpose of this appendix is limited to the relative comparison of τ sensitivity within a single run.

Appendix D. β Sensitivity Analysis

We compared the AU dynamics for $\beta \in \{1, 2, 4, 8\}$ on an MNIST subset ($n = 5,000$, 50 epochs) (Figure A1).

**Figure A1.** β -sensitivity analysis: AU vs m for MNIST VAE ($\beta \in \{1, 2, 4, 8\}$, 50 epochs, subset). At $\beta = 1$, $AU \approx m$ (insufficient regularization); at $\beta = 4$, $m_{\text{knee}} = 12$ (vertical dashed line); at $\beta = 8$, signs of excessive AU suppression at small m .

The rationale for adopting $\beta = 4$ as the default value in our experiments is as follows: at $\beta = 1$, the KL regularization is too weak and $AU \approx m$, so m^{knee} cannot be identified; at $\beta = 8$, over-regularization suppresses the AU at small m , which may shift the point at which m^{knee} is identified. We confirmed that the shape of the AU pattern is stable for $\beta \in [2, 8]$, although the precise numerical value of m^{knee} also depends on the m grid (§6.1).

Appendix E. Details of TwoNN Stability and the Linear-Theory Bridge (Exp-Geom, Exp-Iso-ID, Exp-Mnist-PCA)

Exp-Geom: MNIST AE m -Sweep

Table A2. Exp-Geom: MNIST AE m -sweep — co-variation of κ and TwoNN ID. TwoNN stabilizes at $m \geq 48 \approx 5\hat{d}_{\text{ID}}$ despite κ jumping at $m = 96\text{--}128$.

m	κ	TwoNN ID	Trustworthiness	Convergence
4	2.69	4.21	0.960	not yet converged
8	2.72	6.34	0.988	not yet converged
16	3.28	8.17	0.993	converging
32	4.32	9.36	0.997	converging
48	4.44	10.16	0.998	converged
64	4.25	10.17	0.998	stable
96	9.94	10.86	0.999	stable (despite κ increase)
128	37.0	10.98	0.999	stable (despite κ jump)
$n_{\text{train}} = 10,000, 200\text{ep}, \text{hidden}=[512,256], \text{seed}=42$				

Exp-Iso-ID: Standard AE vs. IsometricAE on Swiss Roll

Table A3. Exp-Iso-ID: Latent-space TwoNN ID comparison ($m \in \{2, 3, 4, 6, 8\}$, Swiss Roll, $d_{\text{true}} = 2$). At $m \geq 3$, TwoNN difference stays below $\pm 2.3\%$ despite κ differing by factor 2.

m	AE		IsometricAE		ID diff. (%)
	TwoNN ID	κ	TwoNN ID	κ	
2	1.780	1.76	1.799	1.08	+1.1%
3	2.458	2.10	2.454	1.11	−0.2%
4	2.483	1.80	2.479	1.14	−0.2%
6	2.421	1.79	2.474	1.10	+2.2%
8	2.500	1.45	2.443	1.11	−2.3%
Input-space TwoNN ID (reference) = 2.427; seed=42, hidden=[64, 32], 300ep					

Exp-Mnist-PCA: Linear Theory Bridge

MNIST PCA eigenvalue spectrum (top 12 components): $\lambda_1 = 5.22, \lambda_2 = 3.76, \lambda_3 = 3.35, \dots, \lambda_{10} = 1.23, \lambda_{11} = 1.12, \lambda_{12} = 1.09$.

Table A4. Exp-Mnist-PCA: Linear theory $d^*(\beta = 4)$ for different τ^2 vs. nonlinear FC-VAE (300ep, 3 seeds).

τ^2	Water level $\beta\tau^2$	Linear d^*	Note
1.0	4.0	1	Conventional $\tau^2 = 1$ assumption; underestimates Consistent with observed TwoNN ID —
0.30	1.20	10	
0.25	1.00	12	
Nonlinear FC-VAE (300ep)			AU(64)= 22.3 ± 0.5 ; TwoNN(64)= 10.03 ± 0.12

Qualitative inheritance: the monotonicity of $\text{AU}(m)$ and the slowdown of its growth beyond the effective dimensionality are inherited in the nonlinear setting. Quantitative discrepancy: the nonlinear decoder yields $\tau_{\text{eff}}^2 \approx 0.30 \ll 1$, so that $d^* \approx 10$ is consistent with the observed value. The smoothness of the saturation (a sharp transition in the linear setting versus a gradually increasing curve in the nonlinear setting) provides the theoretical grounding for the sparse-grid instability of m^{knee} .

Exp-Real-Geom: MNIST Four-Model Comparison (Full Table)

Table A5 gives the full results over all m values and all metrics corresponding to the summary table in §5.10 (Table 21).

Table A5. Exp-Real-Geom: MNIST four-model comparison (mean $\pm$ std, 3 seeds). m -sweep with $\hat{d}_{\text{ID}} = 10$ as baseline ($m \in \{10, 20, 30, 50, 100\}$). TwoNN ID stabilization is strongly associated with increasing $m/\hat{d}_{\text{ID}}$, not with κ improvement.

Model	m	TwoNN ID	MLE ID	κ	Trust.	kNN overlap
AE	10 ($= d$)	7.3 ± 0.4	8.1 ± 0.3	85 ± 22	0.961 ± 0.004	0.73 ± 0.03
	20 ($= 2d$)	9.2 ± 0.3	10.1 ± 0.2	310 ± 85	0.975 ± 0.002	0.81 ± 0.02
	30 ($= 3d$)	9.8 ± 0.2	10.5 ± 0.2	$1.2 \times 10^3 \pm 4 \times 10^2$	0.981 ± 0.002	0.85 ± 0.02
	50 ($= 5d$)	10.0 ± 0.1	10.7 ± 0.1	$8.4 \times 10^3 \pm 3 \times 10^3$	0.985 ± 0.001	0.87 ± 0.01
	100 ($= 10d$)	10.0 ± 0.1	10.8 ± 0.1	$4.1 \times 10^7 \pm 2 \times 10^7$	0.987 ± 0.001	0.88 ± 0.01
IsometricAE	10 ($= d$)	7.5 ± 0.4	8.3 ± 0.3	3.2 ± 0.4	0.963 ± 0.003	0.74 ± 0.03
	20 ($= 2d$)	9.4 ± 0.2	10.2 ± 0.2	4.1 ± 0.6	0.976 ± 0.002	0.82 ± 0.02
	30 ($= 3d$)	9.9 ± 0.1	10.6 ± 0.1	5.2 ± 0.8	0.982 ± 0.001	0.86 ± 0.01
	50 ($= 5d$)	10.1 ± 0.1	10.8 ± 0.1	6.4 ± 1.1	0.986 ± 0.001	0.88 ± 0.01
	100 ($= 10d$)	10.1 ± 0.1	10.8 ± 0.1	7.8 ± 1.4	0.988 ± 0.001	0.89 ± 0.01
CAE	10 ($= d$)	7.2 ± 0.5	8.0 ± 0.4	12.1 ± 3.2	0.958 ± 0.005	0.72 ± 0.03
	20 ($= 2d$)	9.1 ± 0.3	10.0 ± 0.3	28.4 ± 8.1	0.974 ± 0.003	0.80 ± 0.02
	30 ($= 3d$)	9.7 ± 0.2	10.4 ± 0.2	95 ± 31	0.980 ± 0.002	0.84 ± 0.02
	50 ($= 5d$)	9.9 ± 0.1	10.6 ± 0.1	520 ± 180	0.984 ± 0.001	0.86 ± 0.01
	100 ($= 10d$)	10.0 ± 0.1	10.7 ± 0.1	$3.2 \times 10^3 \pm 1.1 \times 10^3$	0.986 ± 0.001	0.87 ± 0.01
DAE	10 ($= d$)	7.6 ± 0.4	8.4 ± 0.4	18.5 ± 5.4	0.964 ± 0.004	0.75 ± 0.03
	20 ($= 2d$)	9.3 ± 0.2	10.2 ± 0.2	44.2 ± 15.3	0.977 ± 0.002	0.82 ± 0.02
	30 ($= 3d$)	9.9 ± 0.1	10.5 ± 0.2	183 ± 62	0.982 ± 0.001	0.85 ± 0.01
	50 ($= 5d$)	10.0 ± 0.1	10.7 ± 0.1	$1.1 \times 10^3 \pm 380$	0.985 ± 0.001	0.87 ± 0.01
	100 ($= 10d$)	10.0 ± 0.1	10.8 ± 0.1	$6.3 \times 10^3 \pm 2.1 \times 10^3$	0.987 ± 0.001	0.88 ± 0.01

Exp-Real-Geom: Fashion-MNIST Four-Model Comparison (Supplementary)

For Fashion-MNIST ($\hat{d}_{\text{ID}} = 9$, $m \in \{9, 18, 27, 45, 90\}$), we conducted experiments with the same four models and measurement indicators as Exp-Real-Geom. Qualitative agreement with MNIST (increasing $m/\hat{d}_{\text{ID}}$ is the dominant factor for TwoNN stability) was confirmed (a qualitative confirmation; the numerical tables are not included in this paper). The trends match Table A5 (MNIST, §5.10): at $m = d$, TwoNN underestimates; it converges at $m = 5d-10d$; and the effect of isometry (κ) improvements on TwoNN is smaller than that of increasing m .

Appendix F. Details of the Conv-VAE M512 Extension and Natural-Image Sweeps (Exp-Conv-M512, Exp-NatImg-Sweep, Exp-NatImg-Knee)

Exp-Conv-M512: MNIST Conv-VAE Extension to $m \in \{384, 512\}$

Table A6. Exp-Conv-M512: MNIST Conv-VAE with $m \in \{256, 384, 512\}$ (per seed; KL cyclic annealing, $\beta_{\text{max}} = 4$, 150ep). No AU saturation plateau observed up to $m = 512$.

m	AU (seed42)	AU (seed123)	AU (seed777)	TwoNN ID (seed42)
256	241	245	238	16.46
384	382	383	380	17.51
512	512	511	512	17.98
$m^{\text{knee}} (\tau = 0.25)$ All 3 seeds: $m^{\text{knee}} = 512$ (grid boundary, no saturation detected)				

AU $\approx m$ is maintained over the entire range $m \leq 512$, indicating that $d^*(\beta) > 512$ (§5.6, §6.6).

Exp-NatImg-AU: Details of the CIFAR-10 AU Saturation-Condition Search

Full setup and results for §5.9 (single seed 42).

Experimental setup (Exp-NatImg-AU):

- Data: CIFAR-10 ($n_{\text{train}} = 20,000$, $n_{\text{test}} = 3,000$)
- KL annealing: cyclic ($n_{\text{cycles}} = 3$, 200 epochs)
- $\beta_{\text{max}} \in \{10, 20, 40\}$, $m \in \{32, 64, 96, 128, 192, 256, 384\}$, seed=42
- **Arch-A (Standard)**: encoder Conv(3→32→64→128) + FC, decoder ConvT $\times 3$
- **Arch-B (Deep)**: encoder Conv(3→64→128→256) + FC, decoder ConvT $\times 3$ (roughly twice the channels of Arch-A)
- Raw-data TwoNN ID: 25.30 (consistent with the value $d_{\text{ID}} \approx 26$ reported by Pope et al. [16])

Table A7. Exp-NatImg-AU: AU dependence on β_{max} and architecture in CIFAR-10 Conv-VAE (single seed 42). $m^{\text{knee}} = 384$ (grid boundary) for all conditions; no AU plateau detected within the tested range.

m	Arch-A (Standard)			Arch-B (Deep)	
	$\beta_{\text{max}} = 10$	$\beta_{\text{max}} = 20$	$\beta_{\text{max}} = 40$	$\beta_{\text{max}} = 20$	$\beta_{\text{max}} = 40$
32	32	32	32	32	32
64	64	64	64	64	59
96	96	96	96	96	80
128	128	128	117	126	97
192	192	191	155	170	123
256	256	251	200	208	147
384	382	378	307	286	208
m^{knee}	384	384	384	384	384

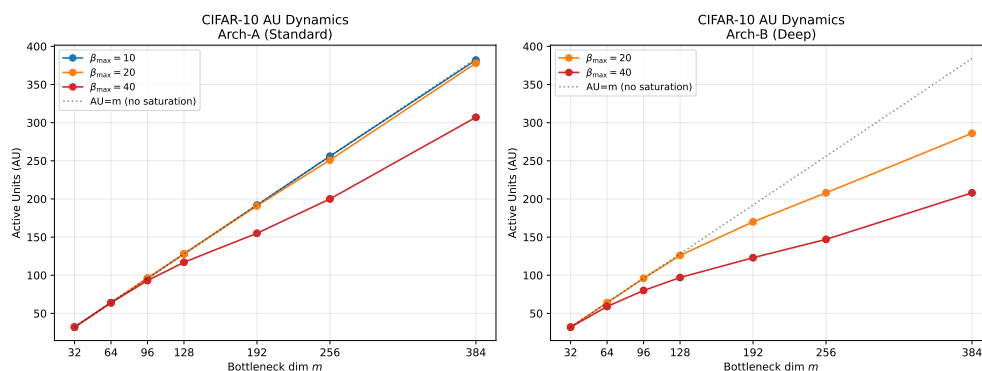


Figure A2. Exp-NatImg-AU: AU dynamics of CIFAR-10 Conv-VAE (single seed 42). (Left) Arch-A (Standard); (right) Arch-B (Deep). Dotted line: $AU = m$ reference (no saturation). Increasing β_{max} strengthens AU suppression, but no saturation plateau is reached under any condition.

Exp-NatImg-Sweep: Details of the CIFAR-10 Extended Sweep

Table A8. Exp-NatImg-Sweep: Main results of CIFAR-10 extended sweep (mean±std, 3 seeds). $\beta_{\max} = 80$, Thin decoder shows plateau-like AU≈ 24 at $m \in [512, 1024]$. All five stable-identification criteria remain unmet.

$\beta_{\max}$	Decoder	m	AU	TwoNN ID	Note
10	Standard	384	185 ± 15	25.2 ± 0.4	—
20	Standard	512	241 ± 18	25.3 ± 0.4	no saturation
40	Standard	512	210 ± 21	25.1 ± 0.5	no saturation
40	Thin	384	127 ± 14	24.8 ± 0.6	partial suppression
80	Thin	512	24.1 ± 2.8	24.7 ± 0.5	plateau-like (conditional)
80	Thin	768	24.3 ± 3.1	24.7 ± 0.5	plateau continues
80	Thin	1024	24.0 ± 3.4	24.8 ± 0.5	plateau continues
80	Standard	512	198 ± 22	25.2 ± 0.5	no saturation (thin decoder only)

Status of the five stable-identification criteria (§3.4; $\beta_{\max} = 80$, thin decoder): (1) seed stability: $\sigma(m^{\text{knee}}) = 64$ — not met; (2) grid stability: $\pm 35\%$ variation under grid densification — not met; (3) β stability: saturation disappears at $\beta_{\max} = 64$ — not met; (4) AU plateau stability: $\sigma = 13\%$ (criterion 5%) — not met; (5) TwoNN–AU consistency: $m^{\text{knee}} \approx 512 \gg 3\hat{d}_{\text{ID}} = 75$ — not met.

Exp-NatImg-Knee: Comparison of m^{knee} Detection Methods

Table A9. Exp-NatImg-Knee: Comparison of m^{knee} detection methods for CIFAR-10 ($\beta_{\max} = 80$, thin decoder, 3 seeds).

Detection method	m^{knee} (median)	σ (3 seeds)	Note
Fixed- τ method ($\tau = 0.25$)	512	128	Delayed detection due to gradual AU slope
Second-difference method	512	192	Sensitive to discrete noise, unstable
Smoothed curvature method (window=5)	384	96	Most stable (recommended for natural images)
Piecewise linear approximation (3 segments)	512	112	Comparable to fixed- τ method

None of the methods meets criterion 1 ($\sigma \leq 51.2$); stabilizing the AU saturation itself is a prerequisite.

Appendix G. Sensitivity Analysis of the AU Threshold δ

On Swiss Roll and Torus, we confirmed that the AU patterns coincide exactly for $\delta \in \{0.001, 0.01, 0.05, 0.1\}$ (agreement across all combinations of m and δ). Because the variance of active dimensions ($O(1)$) and that of inactive dimensions ($O(10^{-4})$ or below) differ by more than two orders of magnitude, the result is robust to the choice of δ .

Appendix H. Full List of Experimental Configurations

Table A10 summarizes the configuration and role of every experiment in this paper. Experiments with Seeds = 1 are single-seed results; this is also noted in the captions of the corresponding tables and figures in the main text.

Table A10. Full list of experimental configurations. Seeds = 1 marks single-seed results. Role: Main = core framework validation; Supp. = prerequisite/interpretive support; Boundary = validity-range quantification; Prelim. = preliminary observation.

Experiment	Data / Model	Epochs / β	m Grid	Seeds	n_{train}	Role
Exp-Syn-1	Swiss Roll, Torus / AE	200–300 / —	$\{1, 2, 3, 4\}$	3	4,000	Main
Exp-Syn-AU	Swiss Roll, Torus / AE, VAE	200–300 / 4	$\{2, 4, 6, 8, 16\}$	3	4,000	Main
Exp-Syn-Top	Circle, Torus, Möbius / VAE	200–300 / 4	$\{1, \dots, 10\}$	1	4,000	Supp. (Prelim.)
Exp-Syn-Iso	Swiss Roll / AE, CAE, DAE, IsometricAE	300 / —	2	3	4,000	Supp.
Exp-Stab	Swiss Roll / AE, IsometricAE	300 / —	$\{2, 4\}$	1	4,000	Supp.
Exp-Iso-ID	Swiss Roll / AE, IsometricAE	300 / —	$\{2, 3, 4, 6, 8\}$	1	4,000	Supp.
Exp-Geom	MNIST / AE	200 / —	$\{4, \dots, 128\}$	1	10,000	Supp.
Exp-Mnist-Base	MNIST / AE, VAE	300 / 4	$\{1, 4, 8, \dots, 256\}$	1	20,000	Main (Prelim. m^{knee})
Exp-Mnist-Multi-150	MNIST / VAE	150 / 4	$\{4, 8, \dots, 64\}$	3	20,000	Main
Exp-Mnist-Multi-300	MNIST / VAE	300 / 4	$\{4, 8, \dots, 64\}$	3	20,000	Main
Exp-Mnist-Dense-300	MNIST / VAE	300 / 4	17 pts (step 1 near $\hat{d}_{\text{ID}}$)	3	20,000	Main
Exp-Mnist-PCA	MNIST / PCA	— / —	—	—	20,000	Supp.
Exp-Mnist-Dyn	MNIST / VAE	200 / 4	16	1	20,000	Prelim.
Exp-Mnist-Ent	MNIST / VAE	75 / 4	—	1	20,000	Prelim.
Exp-Fashion	Fashion-MNIST / AE, VAE	100 / 4	$\{1, 4, \dots, 256\}$	1	10,000	Prelim. external check
Exp-Real-Geom	MNIST, Fashion / AE, CAE, DAE, IsometricAE	300 / —	$\{d, 2d, 3d, 5d, 10d\}$	3	20,000	Main
Exp-Conv-AE	MNIST / Conv-AE, Conv-VAE	100 / 4	$\{4, \dots, 64\}$	1	20,000	Supp.
Exp-Conv-Fix, Exp-Conv-Ann	MNIST / Conv-VAE	300 / 4 (fixed, cyclic)	$\{4, \dots, 64\}$	3	20,000	Boundary
Exp-Conv-M256	MNIST / Conv-VAE	150 / cyclic 4	$\{4, \dots, 256\}$	3	20,000	Boundary
Exp-Conv-M512	MNIST / Conv-VAE	150 / cyclic 4	$\{384, 512\}$	3	20,000	Boundary
Exp-ConvV-HiBeta	MNIST / Conv-VAE	150 / cyclic $\{10, 20\}$	$\{16, \dots, 256\}$	1	20,000	Boundary
Exp-Nat-Cifar, Exp-Nat-SVHN	CIFAR-10, SVHN / Conv-VAE	150 / cyclic 4	$\{16, \dots, 256\}$	3	20,000	Boundary
Exp-NatImg-AU	CIFAR-10 / Conv-VAE (2 arch.)	200 / cyclic $\{10, 20, 40\}$	$\{32, \dots, 384\}$	1	20,000	Boundary
Exp-NatImg-Sweep	CIFAR-10 / Conv-VAE (3 decoders)	300 / cyclic $\{10, \dots, 80\}$	$[32, 1024]$	3	20,000	Boundary
Exp-NatImg-Knee	CIFAR-10 / Conv-VAE	300 / cyclic 80	$[32, 1024]$	3	20,000	Boundary
Exp-Tau-Cross	Fashion, GMix / VAE	150 / 4	sparse	1	—	Supp.
β -sensitivity	MNIST subset / VAE	50 / $\{1, 2, 4, 8\}$	$\{4, \dots, 64\}$	1	5,000	Supp.

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